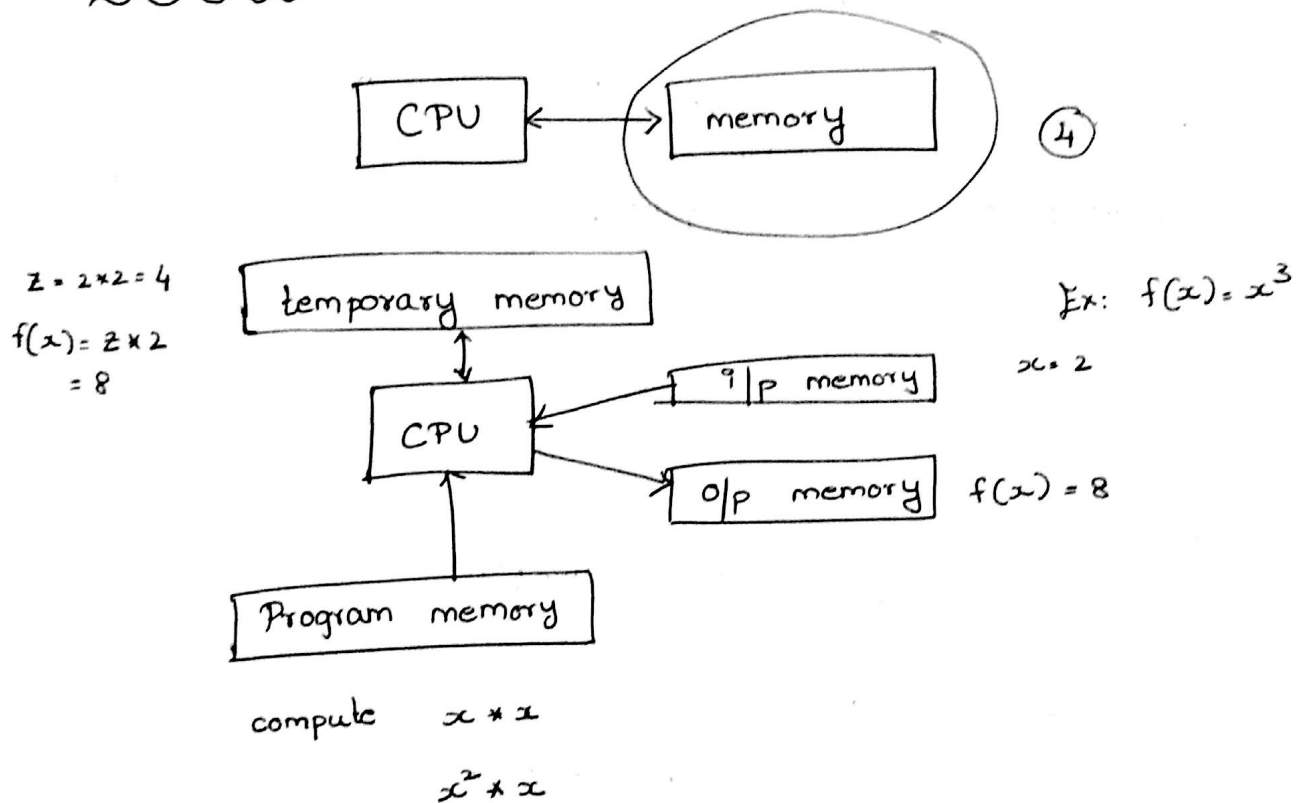
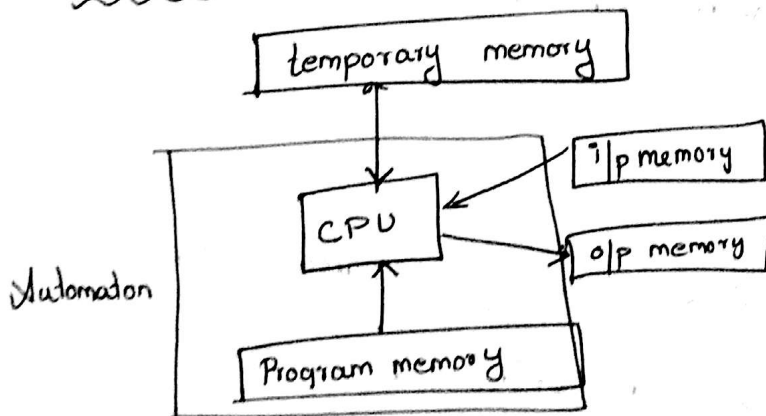


Theory of Computation

Computation:

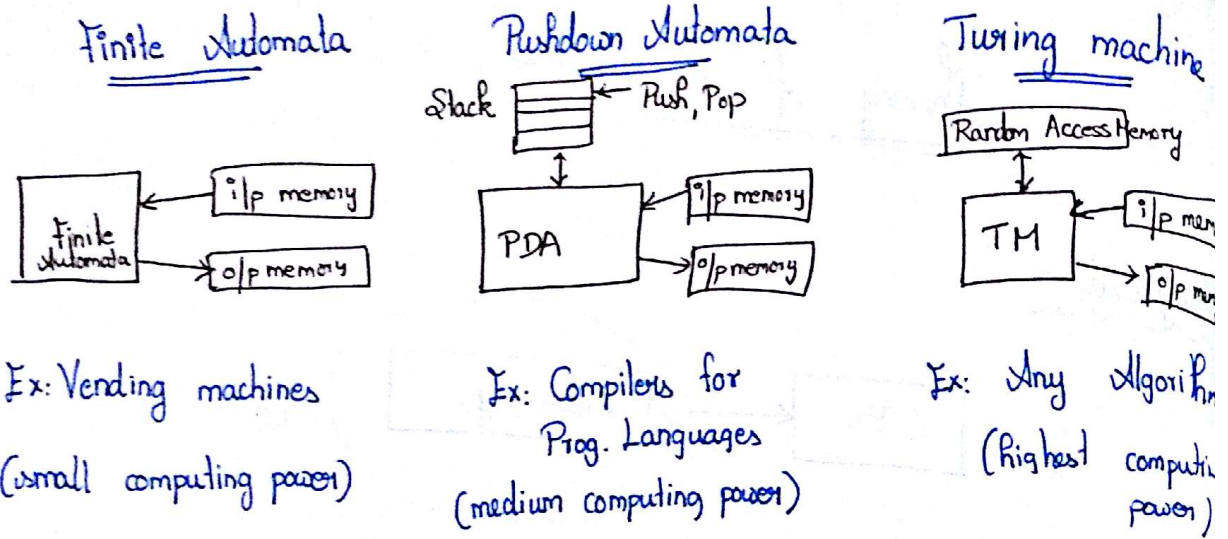


Automaton:



Automata are distinguished by the temporary memory.

- Finite Automata : no temporary memory
- Pushdown Automata : stack
- Turing Machines : random access memory



Central Concepts of Automata Theory:

①. Alphabets:

An alphabet is a finite non-empty set of symbols.
(Usually denoted by Σ).

Ex: $\Sigma = \{0, 1\} \Rightarrow$ binary alphabet.

$\Sigma = \{a, b, \dots, z\} \Rightarrow$ lower-case letters.

Set of all ASCII characters, or the set of all printable ASCII characters.

②. String:

A string over Σ is a sequence of symbols from Σ .

Ex: abbab, baabb are strings over $\Sigma = \{a, b\}$.

③. String Operations:

$w = a_1 a_2 \dots a_n$

$v = b_1 b_2 \dots b_m$

abbab

bbbaaab

Concatenation

$wv = a_1 a_2 \dots a_n b_1 b_2 \dots b_m$

abbabbbbaaab

$$w = a_1 a_2 \dots a_n$$

ababaaaabbb

Reverse:

$$w^R = a_n \dots a_2 a_1$$

bbbbaaababa

String Length:

$$w = a_1 a_2 \dots a_n$$

length: $|w| = n$

$$|abba| = 4$$

$$|aa| = 2$$

$$|a| = 1$$

Length of Concatenation:

$$|u.v| = |u| + |v|$$

$$u = aab, |u| = 3$$

$$v = abaab, |v| = 5$$

$$|u.v| = |aababaab| = 8$$

$$|u.v| = |u| + |v| = 3 + 5 = 8.$$

Empty String:

A string with no letters: λ

$$|\lambda| = 0.$$

$$\lambda w = w \lambda = w.$$

$$\lambda abb = abb \lambda = abb.$$

Substring:

a subsequence of consecutive characters.

String	Substring
abbab	ab
abbab	abba
abbab	b
abbab	bbab

Prefix and Suffix

abbab	
$w = uv$	
↓ prefix	↓ suffix
λ	abbab
a	bbab
ab	bab
abb	ab
abba	b
abbab	λ .

Another Operation:

$$w^n = \underbrace{ww \dots w}_n$$

$$\text{Ex: } (abba)^2 = abbaabba$$

$$w^0 = \lambda$$

$$(abba)^0 = \lambda$$

The * Operation + Operation

Σ^* : the set of all possible strings from alphabet Σ

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{ \lambda, a, b, aa, ab, ba, bb, aaa, aab, \dots \}$$

$$\Sigma^+ = \Sigma^* - \lambda$$

$$\Sigma^+ = \{ a, b, aa, ab, ba, bb, \dots \}$$

Languages:

A language is any subset of Σ^* .

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{ \lambda, a, b, aa, \dots \}$$

Languages: $\{ \lambda \}$

$$\{ a, aa, aab \}$$

$$\{ \lambda, abba, baba, aa, ab, aaaaaa \}$$

Note:

Sets $\phi = \{ \} \neq \{ \lambda \}$

Setsize $|\{ \}| = |\phi| = 0$

$$|\{ \lambda \}| = 1$$

String length $|\lambda| = 0$

An infinite language

$$L = \{ a^n b^n : n \geq 0 \}$$

$$\left. \begin{array}{l} \lambda \\ ab \\ aabb \end{array} \right\}$$

$$\in L$$

$$abb \notin L$$

Operations on Languages:

$$\{a, ab, aaaa\} \cup \{bb, ab\} = \{a, ab, bb, aaaa\}$$

$$\{a, ab, aaaa\} \cap \{bb, ab\} = \{ab\}$$

$$\{a, ab, aaaa\} - \{bb, ab\} = \{a, aaaa\}$$

Complement:

$$\bar{L} = \Sigma^* - L$$

$$\overline{\{a, ba\}} = \{\lambda, b, aa, ab, bb, aaa, \dots\}$$

Reverse:

$$L^R = \{w^R : w \in L\}$$

$$\{ab, aab, baba\}^R = \{ba, baa, abab\}$$

$$L = \{a^n b^n : n \geq 0\}$$

$$L^R = \{b^n a^n : n \geq 0\}$$

Concatenation:

$$L_1 L_2 = \{xy : x \in L_1, y \in L_2\}$$

$$\text{Ex: } \{a, ab, ba\} \{b, aa\}$$

$$= \{ab, aaa, abb, abaa, bab, baaa\}$$

Another Operation:

$$L^n = \underbrace{L L \dots L}_n$$

$$\{a, b\}^3 = \{a, b\} \{a, b\} \{a, b\}$$

$$= \{aaa, aab, aba, abb, baa, bab, bba, bbb\}$$

$$\text{Special case: } L^0 = \{\lambda\}$$

$$\{a, bba, aaa\}^0 = \{\lambda\}$$

Star Closure (Kleene *) :

$$L^* = L^0 \cup L^1 \cup L^2 \dots$$

$$\{a, bb\}^* = \left\{ \begin{array}{l} \lambda, \\ a, bb, \\ aa, abb, bba, bbbb, \dots \\ aaa, aabb, abba, abbbb, \dots \end{array} \right\}$$

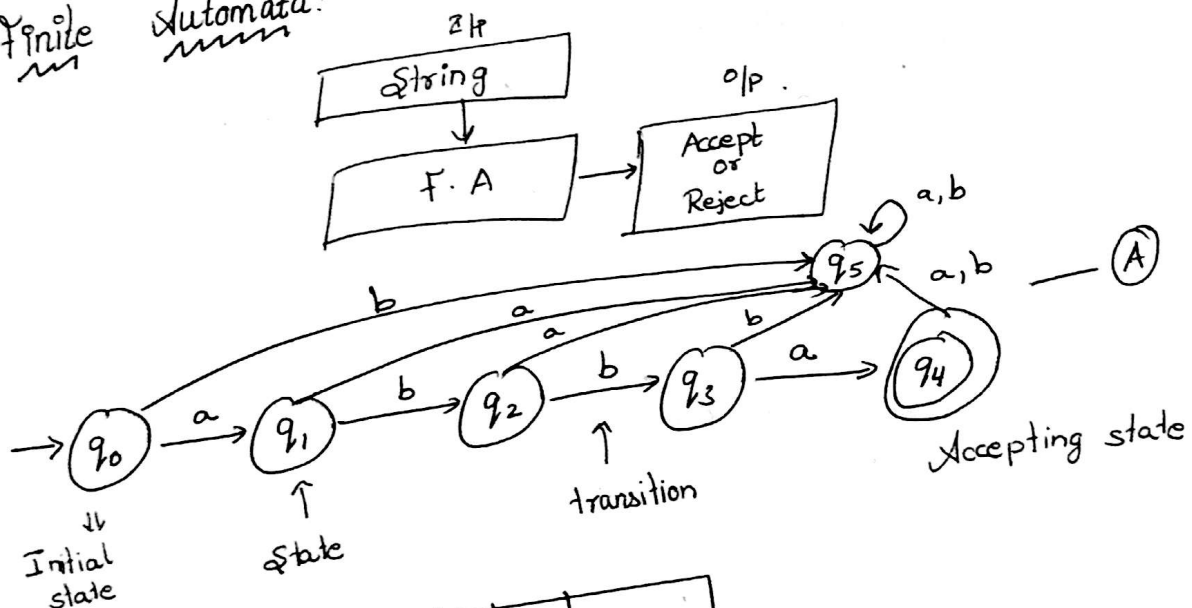
Positive Closure (L^+) :

$$L^+ = L^1 \cup L^2 \cup \dots$$

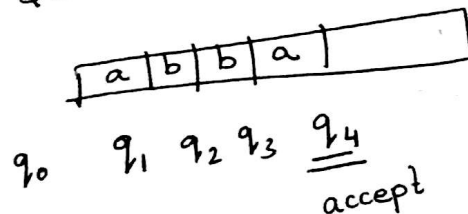
$$= L^* - \{\lambda\}$$

$$\{a, bb\}^+ = \left\{ \begin{array}{l} a, bb \\ aa, abb, bba, bbbb \\ aaa, aabb, abba, abbbb, \dots \end{array} \right\}$$

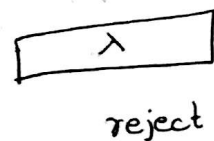
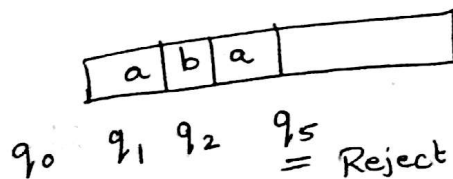
Finite Automata:

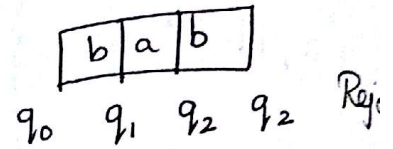
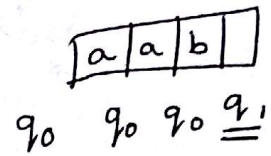
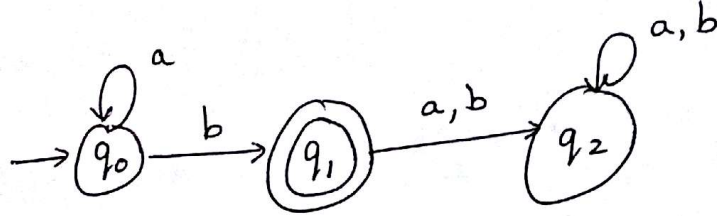


I/P String



Acceptance





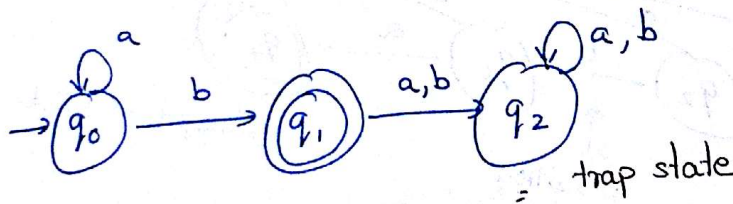
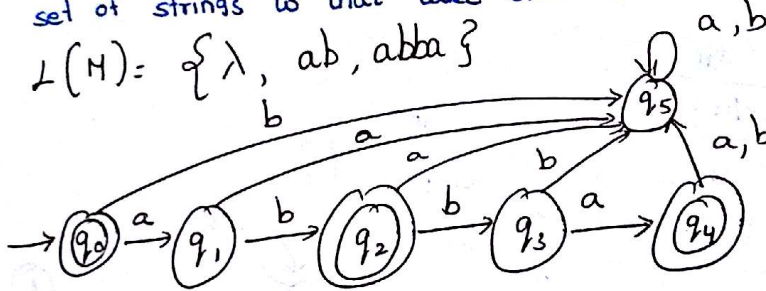
Languages Accepted by DFAs:

The language $L(M)$ contains all i/p strings accepted by M . (Regular Language)

$L(M) = \{ \text{strings that bring } M \text{ to an accepting state} \}$

set of strings w that take start state q_0 to one of the accepting states $\Rightarrow$ the language of A is

$L(M) = \{ \lambda, ab, abba \}$



$L(M) = ?$

$= \{ a^n b : n \geq 0 \}$

FA:
 Deterministic
 Non-deterministic
 Det. Finite Automaton (DFA)
Det. Finite Automaton (DFA)

$M = (Q, \Sigma, \delta, q_0, F)$

$Q \rightarrow$ set of states

$\Sigma \rightarrow$ i/p alphabet

$\delta \rightarrow$ transition f/n.

$q_0 \rightarrow$ initial state

$F \rightarrow$ set of accepting states

Refer (A):

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \{a, b\}$$

$q_0 \rightarrow$ Initial state

$q_4 \rightarrow$ Final state

$$\delta: Q \times \Sigma \rightarrow \begin{cases} Q \\ \subseteq Q \end{cases}$$

single state DFA

set of states NFA

$$\delta(q_0, a) = q_1$$

$$\delta(q_0, b) = q_5$$

Transition Table:

δ	a	b
$\rightarrow q_0$	q_1	q_5
q_1	q_5	q_2
q_2	q_5	q_3
q_3	q_4	q_5
$* q_4$	q_5	q_5
q_5	q_5	q_5

Extended Transition Function δ^* :

$$\delta^*: Q \times \Sigma^* \rightarrow Q$$

①.

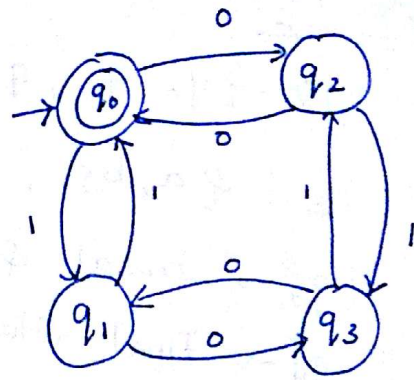
$$\delta^*(q_0, ab) = q_2$$

$$\delta^*(q_0, abba) = q_4$$

$$\delta^*(q_0, abbbbaa) = q_5$$

②.

δ	0	1
* $\rightarrow q_0$	q_2	q_1
q_1	q_3	q_0
q_2	q_0	q_3
q_3	q_1	q_2



$w = 110101$

$$\hat{\delta}(q_0, \epsilon) = q_0$$

$$\hat{\delta}(q_0, 1) = \delta(\hat{\delta}(q_0, \epsilon), 1) = \delta(q_0, 1) = q_1$$

$$\hat{\delta}(q_0, 11) = \delta(\hat{\delta}(q_0, 1), 1) = \delta(q_1, 1) = q_0$$

$$\hat{\delta}(q_0, 110) = \delta(\hat{\delta}(q_1, 1), 0) = \delta(q_0, 0) = q_2$$

$$\hat{\delta}(q_0, 1101) = \delta(\hat{\delta}(q_0, 0), 1) = \delta(q_2, 1) = q_3$$

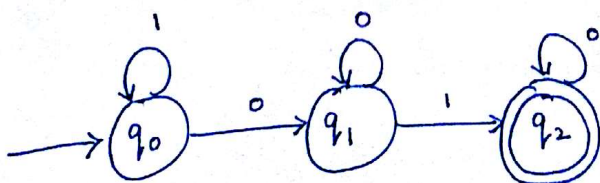
$$\hat{\delta}(q_0, 11010) = \delta(\hat{\delta}(q_2, 1), 0) = \delta(q_3, 0) = q_1$$

$$\hat{\delta}(q_0, 110101) = \delta(\hat{\delta}(q_3, 0), 1) = \delta(q_1, 1) = q_0$$

Construction of DFA:

①. Construct a DFA that accepts all & only the strings of 0's & 1's that have the sequence 01 somewhere in the string.

$$L = \{x01y \mid x \text{ and } y \text{ are any strings of 0's \& 1's}\}$$

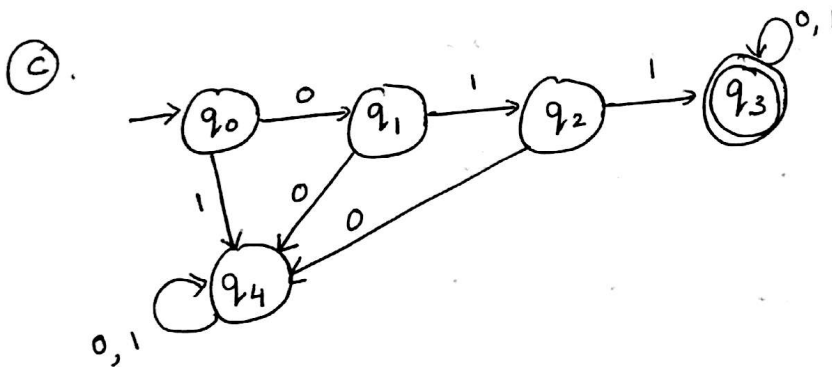
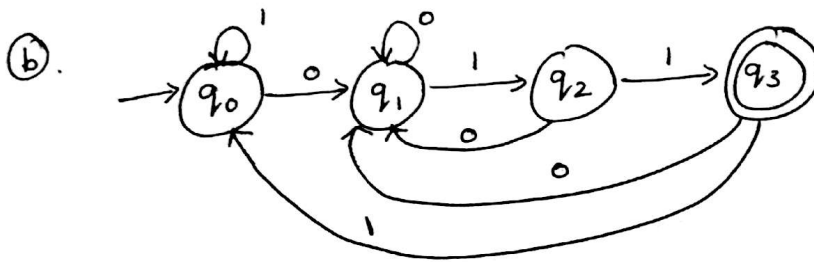
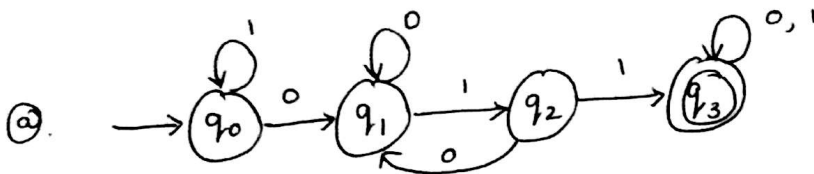


②. Construct a DFA to accept strings over $\{0,1\}$ containing

a. 011 as a substring.

b. 011 as a ending string.

c. 011 as a starting string.



Hint:

- ① Logic
- ② Formal notations
- ③ Transition function (table or diagram)
- ④ Testing the automata

To design any FSM/FA

③ Logic for number divisible by 3

* if we get remainder 0 after dividing a no. by 3
↓
then that no. is divisible by 3

Step 1: $0/3 = 3/3 = 6/3 = 0$ remainder $\rightarrow q_0 \Rightarrow$ consider q_0 as end state since we want remainder

$1/3 = 4/3 = 7/3 = 1$ remainder $\rightarrow q_1$

$2/3 = 5/3 = 8/3 = 2$ remainder $\rightarrow q_2$

Step 2:

$$Q = \{q_0, q_1, q_2\}$$

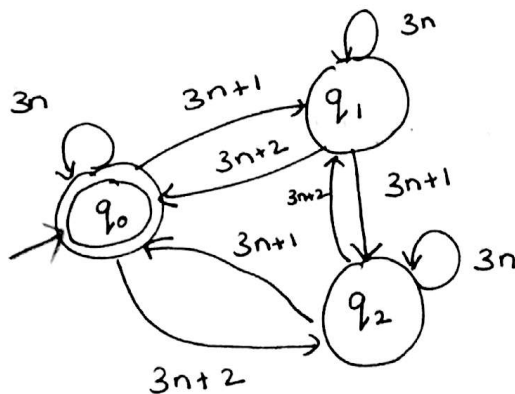
$$\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$\delta = (\Delta) =$ Transition function.

$q_0 =$ start state.

$F = \{q_0\} =$ end state.

	$3n$ 0, 3, 6, 9	$3n+1$ 1, 4, 7	$3n+2$ 2, 5, 8
rem 0 $\rightarrow q_0^*$	q_0	q_1	q_2
rem 1 q_1	q_1	q_2	q_0
rem 2 q_2	q_2	q_0	q_1



Testing the Automata:

i/p string 12345

$$\hat{\delta}(q_0, 12345)$$

$$\hat{\delta}(q_0, 1) = q_1$$

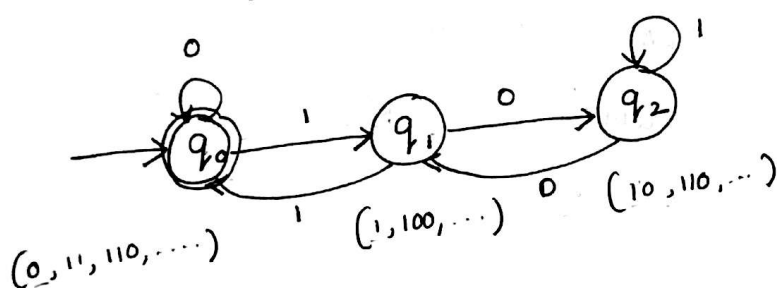
$$\hat{\delta}(q_0, 12) = \delta(\hat{\delta}(q_0, 1), 2) = \delta(q_1, 2) = q_0$$

$$\begin{aligned}\hat{\delta}(q_0, 12, 3) &= \delta(\hat{\delta}(q_0, 2), 3) = \delta(q_0, 3) = q_0 \\ \hat{\delta}(q_0, 1234) &= \delta(\hat{\delta}(q_0, 3), 4) = \delta(q_1, 4) = q_1 \\ \hat{\delta}(q_0, 12345) &= \delta(\hat{\delta}(q_0, 4), 5) = \delta(q_1, 5) = q_0.\end{aligned}$$

Given no is divisible by 3

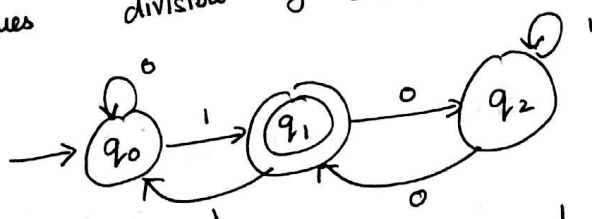
- ④. Construct a minimal DFA which accepts set of all strings over $\{0, 1\}$ which when interpreted as a binary number is divisible by 3.

$$\Sigma = \{0, 1\}$$

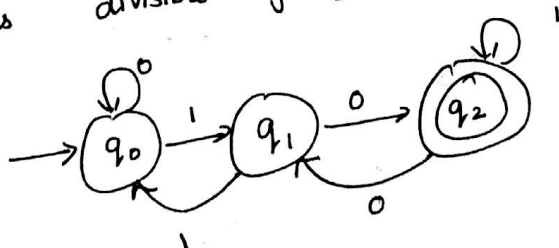


$q_0 \rightarrow \text{rem } 0$
 $q_1 \rightarrow \text{rem } 1$
 $q_2 \rightarrow \text{rem } 2$

- ④ continues divisible by $\equiv 1 \pmod 3$



- ④ continues divisible by $\equiv 2 \pmod 3$



- ④ continues divisible by n
 The No. of states will be n .

	0	1
q_0	q_0	q_1
q_1	q_2	q_0
q_2	q_1	q_2

- ⑤. Construct a minimal DFA which accepts set of all strings over $\{0,1\}$ which when interpreted as a binary number is divisible by 4.

	0	1
$\rightarrow^* q_0$	q_0^*	q_1
q_1	q_2	q_3
q_2	q_0^*	q_1
q_3	q_2	q_3

	0	1
$\rightarrow^* q_0$	q_0	q_1
q_1	q_2	q_1
q_2	q_0	q_1

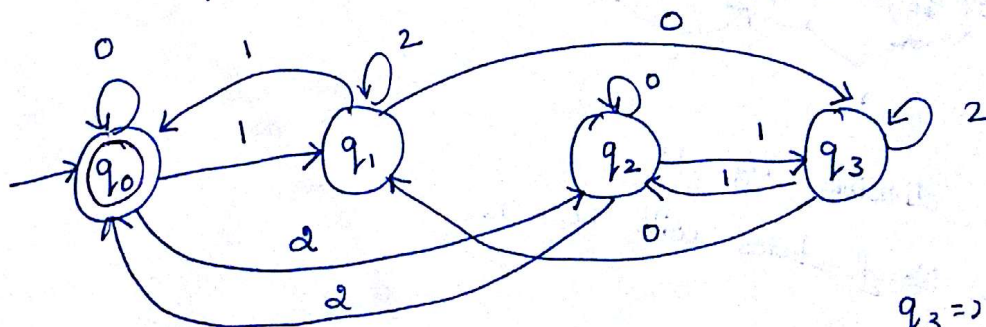
- ⑥. Construct a minimal DFA which accepts set of all strings over $\{0,1,2\}$ which when interpreted as a binary number is divisible by 4.

	0	1	2
$\rightarrow^* q_0$	q_0	q_1	q_2
q_1	q_3	q_0	q_1
q_2	q_2	q_3	q_0
q_3	q_1	q_2	q_3

$$(10)_3 = 3$$

$$(11)_3 = (4)_{10}$$

$$(12)_3 = (5)_{10}$$



$$q_2 \Rightarrow (20)_3 = 6$$

$$(21)_3 = 7$$

$$(22)_3 = 8$$

$$q_3 \Rightarrow (30)_3 = 9$$

$$(31)_3 = 10$$

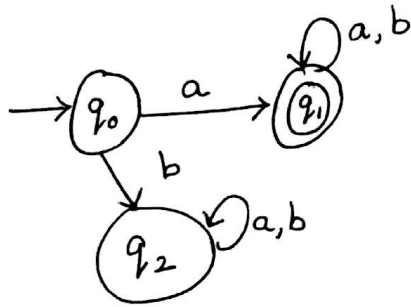
$$(32)_3 = 11$$

Hint:

Base k number $\{0, 1, 2, \dots, k-1\} \Rightarrow$ Alp
 $\Rightarrow$ No. of states
 Divisible by m

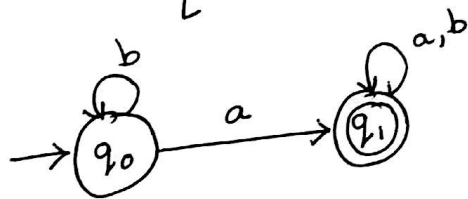
- ⑦. Construct a minimal DFA which accepts set of all strings over $\{a, b\}$ where each string starts with an 'a'.

$$L = \{a, aa, ab, aaa, \dots\}$$



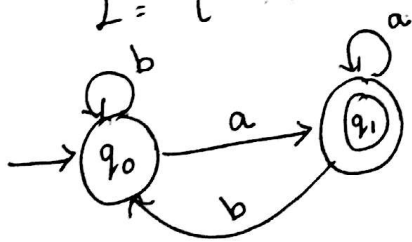
- ⑧. Construct a min. DFA set of all strings containing 'a'.

$$L = \{a, ab, ba, aa, abb, bab, \dots\}$$

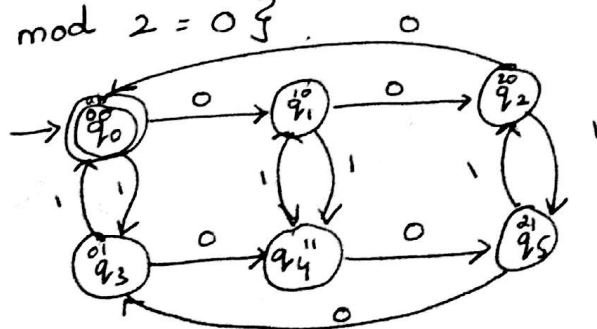


- ⑨. Ending with 'a'.

$$L = \{a, ba, bba, aa, aaa, abaa, \dots\}$$



- ⑩. Construct a DFA to accept strings of 0's & 1's such that $L = \{w \mid w \in (0+1)^* \text{ such that } N_0(w) \bmod 3 = 0 \text{ and } N_1(w) \bmod 2 = 0\}$.

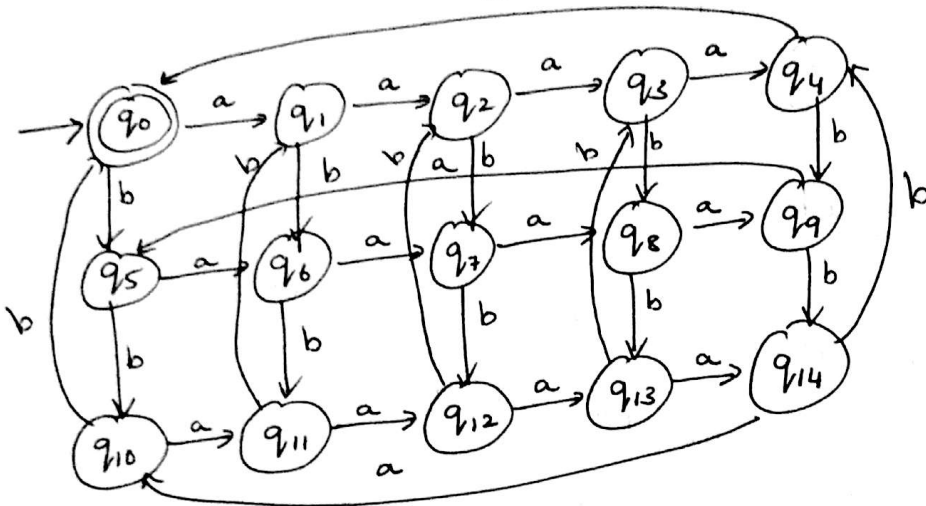


($3 \times 2 = 6$ states)

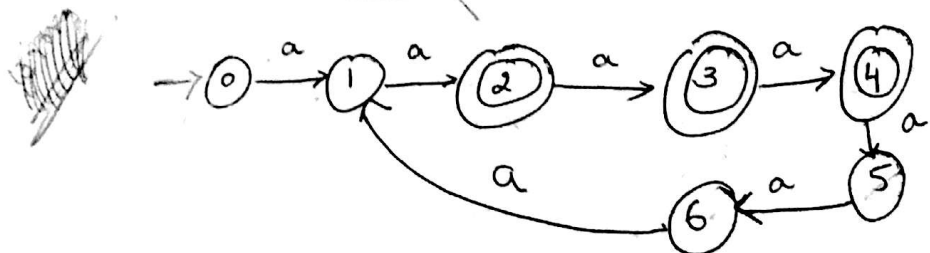
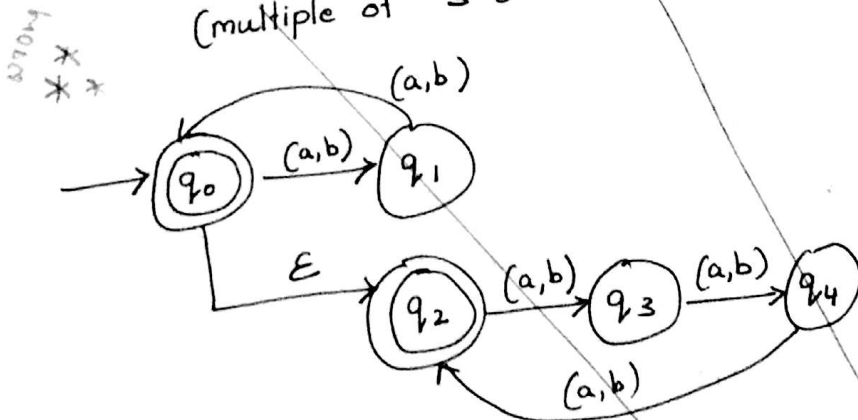
- 11) Construct a DFA to accept strings of a's and b's such that the number of a's is divisible by 5 & number of b's is divisible by 3. (5 x 3 = 15 states)

a) $L = \{w/w \in (a+b)^* \mid N_a(w) \bmod 5 = 0 \text{ \& } N_b(w) \bmod 3 = 0\}$ $\Rightarrow$ Final state q_0

b) $L = \{w/w \in (a+b)^* \mid N_a(w) \bmod 5 = 2 \text{ \& } N_b(w) \bmod 3 = 1\}$ $\Rightarrow$ q_7

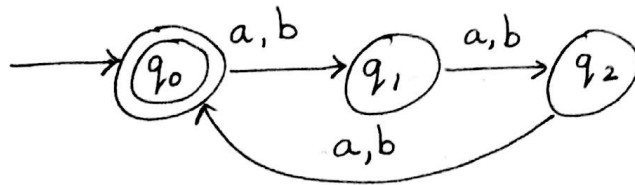


- 12) Construct a DFA accepting the foll. language
 $L = \{w \text{ such that } |w| \bmod 3 \neq |w| \bmod 2\}$
 where $w \in \Sigma = \{a, b\}^*$
 (multiple of 3 or 2 but not both)



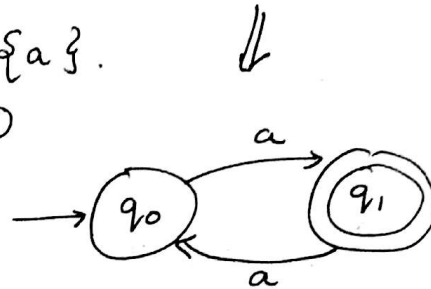
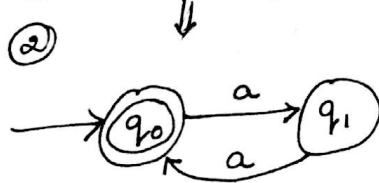
⑬ Design the DFA's for the language

$$L = \{ w : |w| \bmod 3 = 0 \quad w \in (a+b)^* \}$$



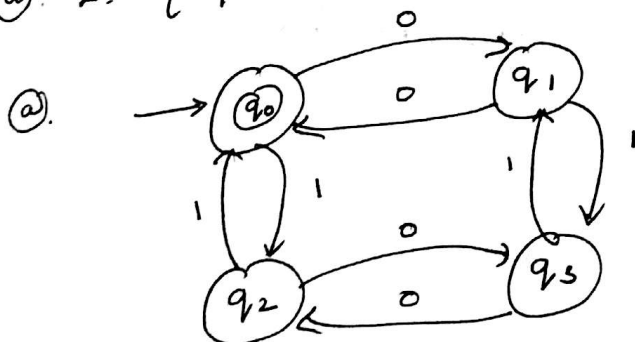
⑭ Construct a DFA to accept ① ~~even~~ ^{odd} no. of a's and

② even no. of a's. $\Sigma = \{a\}$.



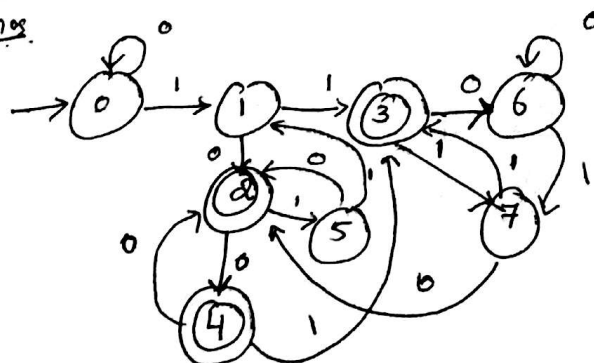
⑮ Construct a DFA to accept

- a. $L = \{ w / w \text{ has both an even no. of 0's \& even no. of 1's} \}$
- b. $L = \{ w / w \text{ has both an even no. of 0's \& odd no. of 1's} \}$
- c. $L = \{ w / w \text{ has both odd no. of 0's \& even no. of 1's} \}$
- d. $L = \{ w / w \text{ has both odd no. of 0's \& odd no. of 1's} \}$



$q_0 \rightarrow$ a
 $q_1 \rightarrow$ c
 $q_2 \rightarrow$ b
 $q_3 \rightarrow$ d

for binary nos.



11
 110
 11001

Nondeterministic Finite Automata: (exhaustive search & backtracking)
 NFA can be defined as $(Q, \Sigma, \delta, q_0, F)$.

$Q \rightarrow$ finite set of states.

$\Sigma \rightarrow$ finite set of i/p symbols.

$q_0 \rightarrow$ member of Q , start state

$F \rightarrow$ subset of Q , is the set of final (or accepting) states

$\delta \rightarrow$ transition function

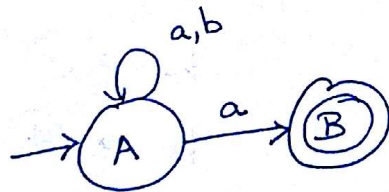
It takes a state in Q and an i/p symbol in Σ as arguments & returns a subset of Q .
 (set of states out of which atleast one state as final)

$$Q \times \Sigma \rightarrow 2^Q \text{ (Power set of } Q \text{)}$$

$$Q = \{q_0, q_1\}$$

$$\Sigma = \{a, b\}$$

$$L_1 = \{\text{strings ends with 'a'}\}$$



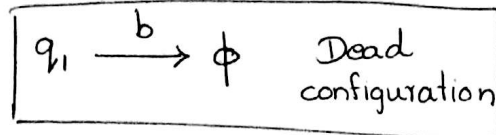
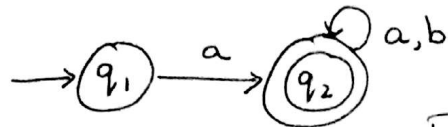
$$A \xrightarrow{a} A, B$$

$$A \xrightarrow{b} A$$

$$B \xrightarrow{a} \phi$$

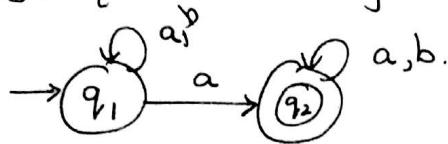
$$B \xrightarrow{b} \phi$$

②. $L_2 = \{ \text{starts with 'a'} \}$ $\Sigma = \{a, b\}$

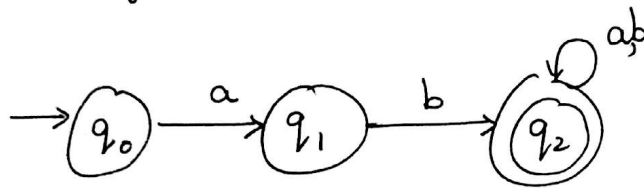


In DFA we used trap state

③. $L_3 = \{ \text{contains 'a'} \}$



④. $L_4 = \{ \text{starts with 'ab'} \}$



abaab

$$\hat{\delta}(q_0, \epsilon) = q_0$$

$$\hat{\delta}(q_0, a) = q_1$$

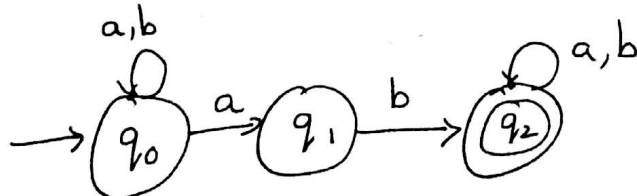
$$\hat{\delta}(q_0, ab) = \delta(q_1, b) = q_2$$

$$\hat{\delta}(q_0, aba) = \delta(q_2, a) = q_2$$

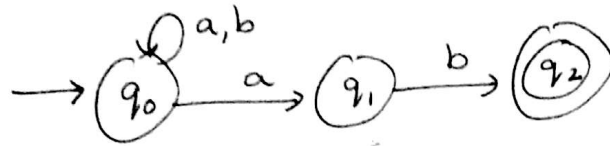
$$\hat{\delta}(q_0, abaa) = \delta(q_2, a) = q_2$$

$$\hat{\delta}(q_0, abaab) = \delta(q_2, b) = \underline{q_2} \text{ accepting state}$$

⑤. $L_5 = \{ \text{contains 'ab'} \}$



⑥. $L_6 = \{ \text{strings ending with 'ab'} \}$



aabab

$$\hat{\delta}(q_0, \epsilon) = q_0$$

$$\hat{\delta}(q_0, a) = \delta(q_0, a) = \{q_0, q_1\}$$

$$\hat{\delta}(q_0, ab) = \delta(q_0, a) \cup \delta(q_1, b) = \{q_0, q_1\} \cup \{q_2\}$$

$$\hat{\delta}(q_0, aab) = \delta(q_0, b) \cup \delta(q_1, b) = \{q_0\} \cup \{q_2\} = \{q_0, q_2\}$$

$$\hat{\delta}(q_0, aaba) = \delta(q_0, a) \cup \delta(q_2, a) = \{q_0, q_1\} \cup \emptyset = \{q_0, q_1\}$$

$$\hat{\delta}(q_0, aabab) = \delta(q_0, b) \cup \delta(q_1, b) = \{q_0\} \cup \{q_2\} = \{q_0, \underline{q_2}\}$$

accepting state

Language of an NFA:

If $A = (Q, \Sigma, \delta, q_0, F)$ is an NFA, then

$$L(A) = \{ w \mid \hat{\delta}(q_0, w) \cap F \neq \emptyset \}$$

$L(A)$ is the set of strings w in Σ^* such that

$\hat{\delta}(q_0, w)$ contains at least one accepting state

Equivalence of Deterministic and Non-deterministic Finite Automata

Which is powerful?

(Both DFA & NFA are equally powerful).

Conversion of $NFA \rightarrow DFA \Rightarrow$ one of the ways
Subset construction method.



NFA

DFA

$$N = (Q_N, \Sigma, \delta_N, q_0, F_N) \xRightarrow{\text{Goal}} D = (Q_D, \Sigma, \delta_D, q_0, F_D)$$

such that $L(D) = L(N)$.

$Q_D \rightarrow$ set of subsets of Q_N



if n states then Q_D will have
worst case 2^n states.

$F_D \rightarrow$ is all sets of N 's states that include
at least one accepting state of N .

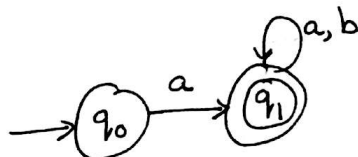
$S \Rightarrow$ set of
subsets.

For each set $S \subseteq Q_N$ and for each i/p symbol $a \in \Sigma$,

$$\delta_D(S, a) = \bigcup_{p \in S} \delta_N(p, a)$$

①. $L_1 = \{ \text{starts with 'a'} \}$. $\Sigma = \{a, b\}$

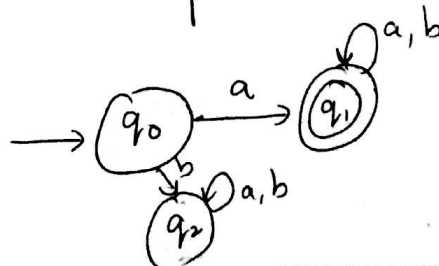
NFA



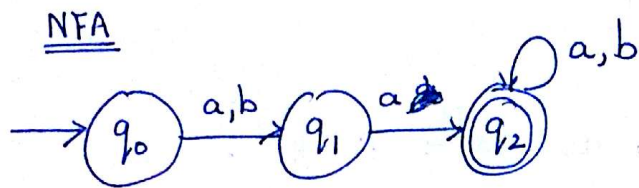
DFA

	a	b
$\rightarrow q_0$	q_1	q_2
q_1^*	q_1	q_1
trap state q_2	q_2	q_2

	a	b
$\rightarrow q_0$	q_1	ϕ
$* q_1$	q_1	q_1



②. $L_2 = \{ \text{all strings in which second symbol from RHS is 'a'} \}$ $\Sigma = \{a, b\}$



	a	b
q_0	$\{q_1\}$	$\{q_1\}$
q_1	$\{q_2\}$	ϕ
q_2^*	$\{q_2\}$	$\{q_2\}$

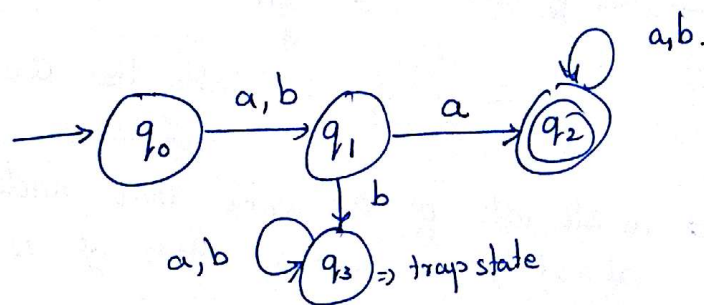
DFA

	a	b
q_0	q_1	q_1
q_1	q_2	q_3
q_2^*	q_2	q_2
q_3	q_3	q_3

$$\delta(q_0, a) = q_1$$

$$\delta(q_0, b) = q_1$$

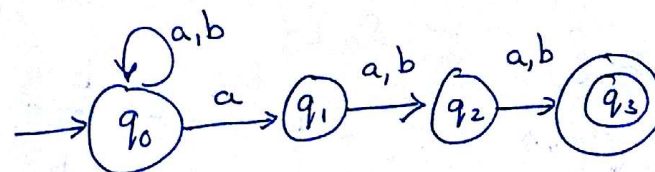
$$\delta(q_1, a) = q_2$$



③. $L_3 = \{ \text{strings in which 3rd symbol from RHS is 'a'} \}$ $\Sigma = \{a, b\}$

$$L = \{ \boxed{\dots} a \boxed{\dots}, \dots \}$$

NFA



	a	b
q_0	$\{q_0, q_1\}$	q_0
q_1	$\{q_2\}$	$\{q_2\}$
q_2	$\{q_3\}$	$\{q_3\}$
q_3^*	ϕ	ϕ

Lazy Evaluation
 ①. q_0 on 'a' $\{q_0, q_1\}$

$$\begin{aligned} \textcircled{2} \quad \delta(\{q_0, q_1\}, a) &= \{q_0, q_1\} \cup q_2 ; \delta(\{q_0, q_1\}, b) = \\ &= \{q_0, q_1, q_2\} \quad \{q_0, q_2\} \end{aligned}$$

q_0 on 'b' $\{q_0\}$

③. q_1 on 'a' $\{q_2\}$; q_1 on 'b' $\{q_2\}$

④. q_2 on 'a' $\{q_3\}$; q_2 on 'b' $\{q_3\}$

⑤. $\{q_0, q_1, q_2\}$ on 'a' & 'b'

$$\delta(\{q_0, q_1, q_2\}, a) = \{q_0, q_1, q_2, q_3\}$$

$$\delta(\{q_0, q_1, q_2\}, b) = \{q_0, q_2, q_3\}$$

⑥. $\{q_0, q_2, q_3\}$ on 'a' & 'b'

$$\delta(\{q_0, q_2, q_3\}, a) = \{q_0, q_1, q_3\}$$

$$\delta(\{q_0, q_2, q_3\}, b) = \{q_0, q_3\}$$

⑦. $\{q_0, q_2\}$ on 'a' & 'b'

$$\delta(\{q_0, q_2\}, a) = \{q_0, q_1, q_3\}$$

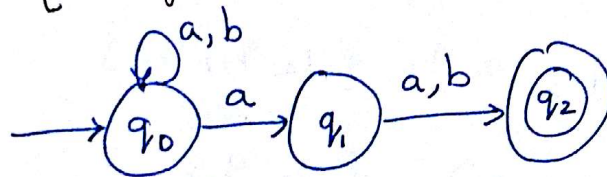
$$\delta(\{q_0, q_2\}, b) = \{q_0, q_3\}$$

⑧. $\{q_0, q_3\}$ on 'a' & 'b'

$$\delta(\{q_0, q_3\}, a) = \{q_0, q_1\} ; \delta(\{q_0, q_3\}, b) = \{q_0\}$$

$\rightarrow q_0$	$\{q_0, q_1\}$	q_0
$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_2\}$
$\{q_0, q_2\}$	$\{q_0, q_1, q_3\}$	$\{q_0, q_3\}$
$\{q_0, q_3\}^*$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_3\}$
$\{q_0, q_1, q_3\}^*$	$\{q_0, q_1, q_2\}$	$\{q_0, q_2\}$
$\{q_0, q_1, q_2, q_3\}^*$	$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_2, q_3\}$
$\{q_0, q_2, q_3\}^*$	$\{q_0, q_1, q_3\}$	$\{q_0, q_3\}$

(4). $L_4 = \{ \text{strings in which second symbol from RHS is "a"} \}$



NFA

	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	q_0
q_1	q_2	q_2
q_2^*	ϕ	ϕ

DFA

	a	b
q_0	$\{q_0, q_1\}$	q_0
q_1	q_2	q_2
$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_2\}$
$\{q_0, q_2\}^*$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_0, q_1, q_2\}^*$	$\{q_0, q_1, q_2\}$	$\{q_0, q_2\}$

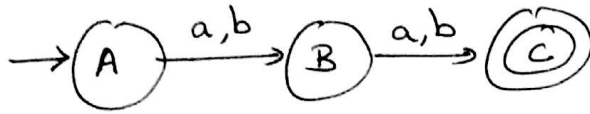
* Problems on NFA $L = \{ \text{strings of length } 2 \}$ $\Sigma = \{a, b\}$

(a) exactly 2

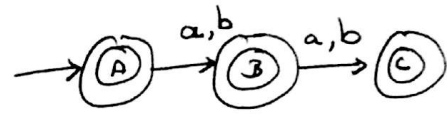
(b) at most 2

(c) at least 2

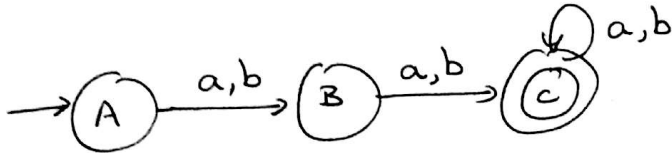
(a). $L = \{aa, ab, ba, bb\}$



(b). $L = \{\epsilon, a, b, aa, ab, ba, bb\}$



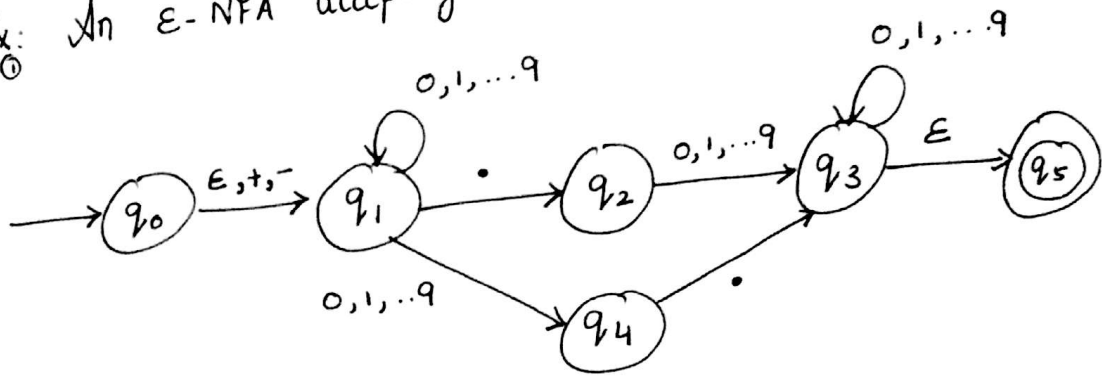
(c). $L = \{aa, ab, ba, bb, aa, \dots\}$



Complementing NFA and DFA:

Epsilon NFA:

Ex: An ϵ -NFA accepting decimal numbers.



	ϵ	$+, -$	$.$	$0, 1, \dots, 9$
$\rightarrow q_0$	$\{q_1\}$	$\{q_1\}$	$\emptyset$	$\emptyset$
q_1	$\emptyset$	$\emptyset$	$\{q_2\}$	$\{q_1, q_4\}$
q_2	$\emptyset$	$\emptyset$	$\emptyset$	$\{q_3\}$
q_3	$\{q_5\}$	$\emptyset$	$\emptyset$	$\{q_3\}$
q_4	$\emptyset$	$\emptyset$	$\{q_3\}$	$\emptyset$
q_5	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

Language:

$$L(E) = \{w \mid \hat{\sigma}(q_0, w) \cap F \neq \emptyset\}$$

$$E = (Q, \Sigma, \delta, q_0, F)$$

$$\downarrow$$

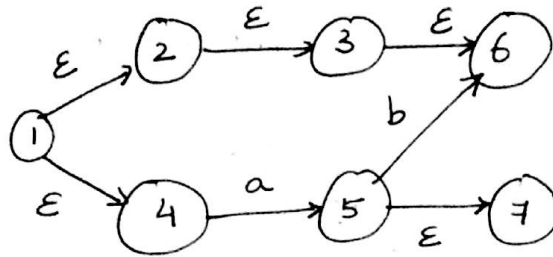
$$\Sigma \cup \{\epsilon\}$$

$$\delta: Q \times \Sigma \cup \{\epsilon\} \rightarrow 2^Q$$

$$E_{CLOSE}(q_0) = \{q_0, q_1\}$$

$$E_{CLOSE}(q_3) = \{q_3, q_5\}$$

Ex(2):



$$\text{ECLOSE}(1) = \{1, 2, 3, 6, 4\}$$

Extended Transition function:

5.67

$$\begin{aligned} \hat{\delta}(q_0, \epsilon) &= \text{ECLOSE}(q_0) = \{q_0, q_1\} \\ \hat{\delta}(q_0, 5) &= \hat{\delta}(\{q_0, q_1\}, 5) = \delta(q_0, 5) \cup \delta(q_1, 5) \\ &= \phi \cup \{q_1, q_4\} \\ \hat{\delta}(q_0, 5 \cdot) &= \hat{\delta}(\{q_1, q_4\}, 5 \cdot) = \delta(q_1, \cdot) \cup \delta(q_4, \cdot) \\ &= \{q_2\} \cup \{q_3, q_5\} \\ &= \{q_2, q_3, q_5\} \\ \hat{\delta}(q_0, 5 \cdot 6) &= \hat{\delta}(\{q_2, q_3, q_5\}, 5 \cdot 6) = \delta(q_2, 6) \cup \delta(q_3, 6) \cup \delta(q_5, 6) \\ &= \{q_3\} \cup \{q_3, q_5\} \cup \phi \end{aligned}$$

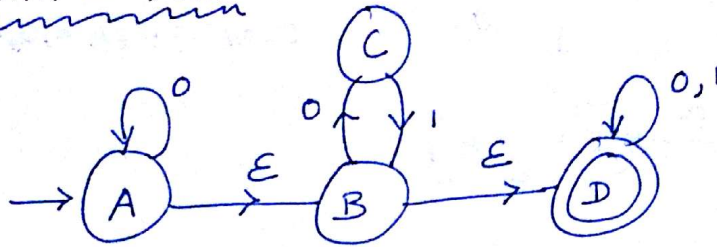
5.678

	ECLOSE	Σ
		5.678
$\text{ECLOSE}(q_0)$	$\{q_0, q_1\}$	⑤
$\text{ECLOSE}(q_1, q_4)$	$\{q_1, q_4\}$	⑥
$\text{ECLOSE}(q_2, q_3)$	$\{q_2, q_3, q_5\}$	⑦
$\text{ECLOSE}(q_3)$	$\{q_3, q_5\}$	⑧
$\text{ECLOSE}(q_5)$	$\{q_3, q_5\}$	$\{q_3\}$
$\text{ECLOSE}(q_3)$	$\{q_3, q_5\}$	$\{q_3\}$
$\text{ECLOSE}(q_5)$	$\{q_3, q_5\}$	accepting state.

E-NFA $\rightarrow$ NFA

DFA $\cong$ NFA $\cong$ E-NFA

(1)



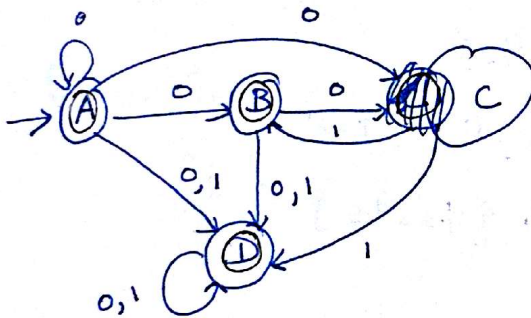
NFA

	0	1
$\rightarrow A$	A, B, C, D	D
B	C, D	D
C	ϕ	B, D
D^*	D	D

T.F for E-NFA $\rightarrow$ NFA

ϵ -closure ($\delta(\epsilon$ -closure(A), 0))

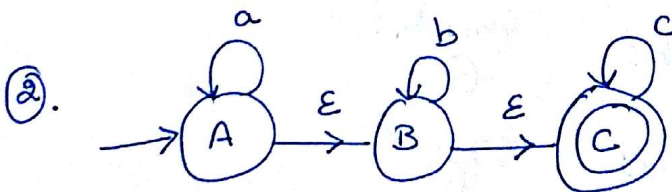
	ϵ^*	0	ϵ^*
A	A, B, D	A, C, D	A, B, C, D
B	B, D	C, D	C, D
C	C	ϕ	ϕ
D	D	D	D



Final States A, B, D

ϵ -closure ($\delta(\epsilon$ -closure(A), 1))

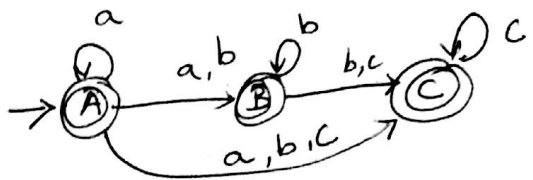
	ϵ^*	1	ϵ^*
A	A, B, D	D	D
B	B, D	D	D
C	C	B	B, D
D	D	D	D



NFA

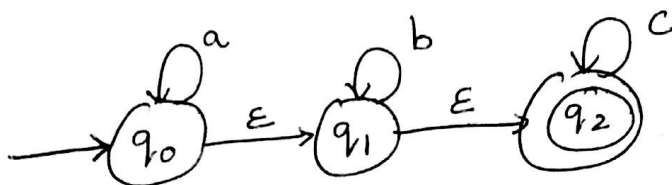
	a	b	c
$\rightarrow A$	{A, B, C}	{B, C}	{C}
B	ϕ	{B, C}	{C}
C^*	ϕ	ϕ	{C}

	ϵ^*	a	ϵ^*
A	A, B, C	A	A, B, C
B	B, C	ϕ	ϕ
C	C	ϕ	ϕ
	ϵ^*	b	ϵ^*
A	A, B, C	B	B, C
B	B, C	B	B, C
C	C	ϕ	ϕ



E-NFA $\xrightarrow{\text{min}}$ DFA

①.



Step 1:

Eclosure of initial state

And then follow subset construction.

$$\text{Eclosure}(q_0) = \{q_0, q_1, q_2\} \text{ --- (A)}$$

(A) on a

$$\delta(\{q_0, q_1, q_2\}, a) = q_0$$

(A) on b

$$\delta(\{q_0, q_1, q_2\}, b) = q_1$$

(A) on c

$$\delta(\{q_0, q_1, q_2\}, c) = q_2$$

(B) on a

$$\delta(\{q_1, q_2\}, a) = \phi$$

(B) on b

$$\delta(\{q_1, q_2\}, b) = q_1$$

(B) on c

$$\delta(\{q_1, q_2\}, c) = q_2$$

(C) on a

(C) on b

(C) on c

on Eclosure

$$\text{Eclosure}(q_0) = \{q_0, q_1, q_2\} \text{ (A)}$$

on Eclosure

$$\text{Eclosure}(q_1) = \{q_1, q_2\} \text{ (B)}$$

on Eclosure

$$\text{Eclosure}(q_2) = \{q_2\} \text{ (C)}$$

on Eclosure

ϕ

on Eclosure

$$\text{Eclosure}(q_2) = \{q_1, q_2\} \text{ (B)}$$

on Eclosure

$$\text{Eclosure}(q_2) = \{q_2\} \text{ (C)}$$

$$\text{E.closure}(\phi) = \phi$$

$$\text{E.closure}(\phi) = \phi$$

$$\text{E.closure} \phi = C$$

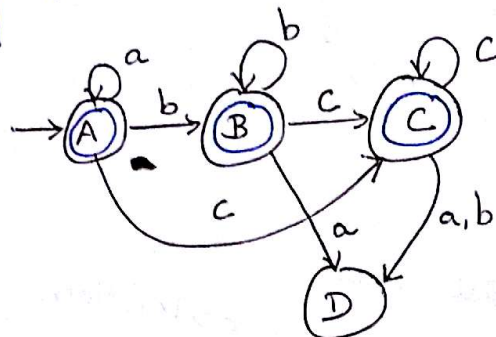
	a	b	c
A	A	B	C
B	ϕ	B	C
C	ϕ	ϕ	C

$$A \rightarrow q_0, q_1, q_2$$

$$B \rightarrow q_1, q_2$$

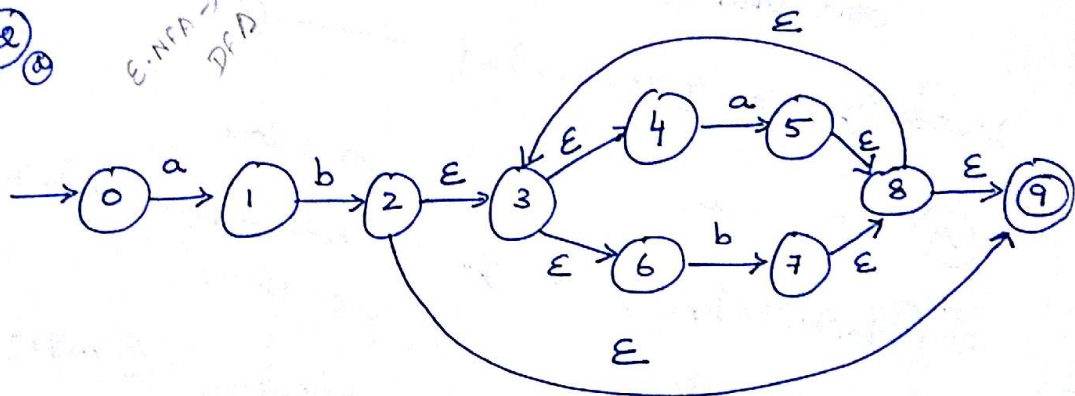
$$C \rightarrow q_2$$

$$D \rightarrow \phi$$



②

E-NFA $\rightarrow$ DFA



$$E\text{-close}(0) = 0 \quad \textcircled{A}$$

A on a

E-close

$$\delta(A, a) = 1$$

$$E\text{-close}(1) = 1 \quad \textcircled{B}$$

A on b

$$\delta(A, b) = \phi$$

$$E\text{-close}(\phi) = \phi$$

$$E\text{-close}(\phi) = \phi$$

B on a $\delta(B, a) = \phi$

B on b $\delta(B, b) = 2$

$$E\text{-close}(2) = \{2, 3, 4, 6, 9\} \quad \textcircled{C}$$

C on a $\delta(C, a) = 5$

$$E\text{-close}(5) = \{5, 8, 3, 4, 6, 9\} \quad \textcircled{D}$$

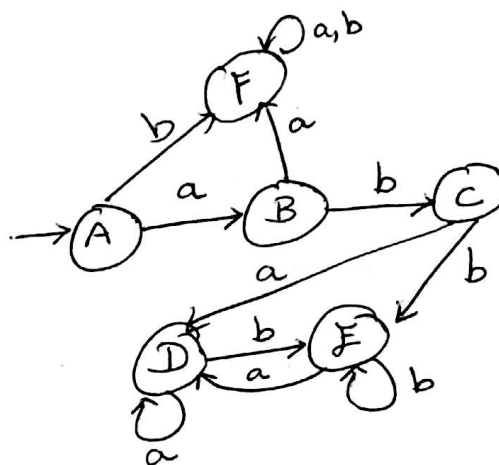
C on b $\delta(C, b) = 7$

$$E\text{-close}(7) = \{7, 8, 9, 3, 4, 6\} \quad \textcircled{E}$$

$\textcircled{D}$ on a $\delta(\textcircled{D}, a) = 5$
 $\textcircled{D}$ on b $\delta(\textcircled{D}, b) = 7$
 $\textcircled{E}$ on a $\delta(\textcircled{E}, a) = 5$
 $\textcircled{E}$ on b $\delta(\textcircled{E}, b) = 7$

$\epsilon\text{-close}(5) = \{5, 8, 3, 4, 6, 9\} \textcircled{D}$
 $\epsilon\text{-close}(7) = \{7, 8, 9, 3, 4, 6\} \textcircled{E}$
 $\epsilon\text{-close}(5) = \{5, 8, 3, 4, 6, 9\} \textcircled{D}$
 $\epsilon\text{-close}(7) = \{7, 8, 9, 3, 4, 6\} \textcircled{E}$

δ	a	b
$\rightarrow A$	B	ϕ
B	ϕ	C
C^*	D	E
D^*	D	E
E^*	D	E



$\textcircled{b}$ E-NFA to NFA

$$\delta(\epsilon\text{-closure}(0), a) = \epsilon\text{-closure}(1) = 1$$

$$\delta(\epsilon\text{-closure}(0), b) = \epsilon\text{-closure}(\phi) = \phi$$

$$\delta(\epsilon\text{-closure}(1), a) = \epsilon\text{-closure}(\phi) = \phi$$

$$\delta(\epsilon\text{-closure}(1), b) = \epsilon\text{-closure}(2) = \{2, 3, 4, 6, 9\}$$

$$\delta(\epsilon\text{-closure}(2), a) = \epsilon\text{-closure}(5) = \{5, 8, 3, 4, 6, 9\}$$

$$\delta(\epsilon\text{-closure}(2), b) = \epsilon\text{-closure}(7) = \{7, 8, 3, 4, 6, 9\}$$

$$\delta(\epsilon\text{-closure}(3), a) = \epsilon\text{-closure}(5) = \{5, 8, 3, 4, 6, 9\}$$

$$\delta(\epsilon\text{-closure}(3), b) = \epsilon\text{-closure}(7) = \{7, 8, 3, 4, 6, 9\}$$

$$\delta(\epsilon\text{-closure}(4), a) = \epsilon\text{-closure}(5) = \{5, 8, 3, 4, 6, 9\}$$

$$\delta(\epsilon\text{-closure}(4), b) = \epsilon\text{-closure}(\phi) = \phi$$

$$\delta(\epsilon\text{-closure}(5), a) = \epsilon\text{-closure}(5) =$$

$$\delta(\epsilon\text{-closure}(5), b) = \epsilon\text{-closure}(7) =$$

δ	a	b
$\rightarrow 0$	1	ϕ
1	ϕ	$\{2, 3, 4, 6, 9\}$
2^*	$\{5, 8, 3, 4, 6, 9\}$	$\{7, 8, 3, 4, 6, 9\}$
3	$\{5, 8, 3, 4, 6, 9\}$	$\{7, 8, 3, 4, 6, 9\}$
4	$\{5, 8, 3, 4, 6, 9\}$	ϕ
5^*	$\{5, 8, 3, 4, 6, 9\}$	$\{7, 8, 3, 4, 6, 9\}$
6	ϕ	$\{7, 8, 3, 4, 6, 9\}$
7^*	$\{5, 8, 3, 4, 6, 9\}$	$\{7, 8, 3, 4, 6, 9\}$
8^*	$\{5, 8, 3, 4, 6, 9\}$	$\{7, 8, 3, 4, 6, 9\}$
9^*	ϕ	ϕ

$$\begin{aligned}
 \delta(\text{e-closure}(6), a) &= \text{e-closure}(\phi) = \phi \\
 \delta(\text{e-closure}(6), b) &= \text{e-closure}(7) = \\
 \delta(\text{e-closure}(7), a) &= \text{e-closure}(5) = \\
 \delta(\text{e-closure}(7), b) &= \text{e-closure}(7) = \\
 \delta(\text{e-closure}(8), a) &= \text{e-closure}(5) = \\
 \delta(\text{e-closure}(8), b) &= \text{e-closure}(7) = \\
 \delta(\text{e-closure}(9), a) &= \text{e-closure}(\phi) = \\
 \delta(\text{e-closure}(9), b) &= \text{e-closure}(\phi) =
 \end{aligned}$$

(
Equivalence & Minimization of Automata:

Equivalence of 2 States:

① Equivalent (Indistinguishable):

$$\begin{aligned}
 &\delta(p, w) \in F \quad \& \quad \delta(q, w) \in F \\
 + \\
 &\delta(p, w) \notin F \quad \& \quad \delta(q, w) \notin F
 \end{aligned}$$

② Distinguishable:

$$\delta(p, w) \in F \quad \& \quad \delta(q, w) \notin F.$$

B.

Bc

C.

C

Partition Method for minimization of states of DFA

Step 1: Remove all unreachable states from the start state.

Step 2: Construct Π_{old} : Partition the states into 2 groups one consisting of final states and other consisting of non-final states.

Step 3: Construct Π_{new} from Π_{old} .

[Find $\delta(q, a)$ for each q in G on i/p symbol]

If transition results with the states in different groups proceed with partition

else no need for partition.

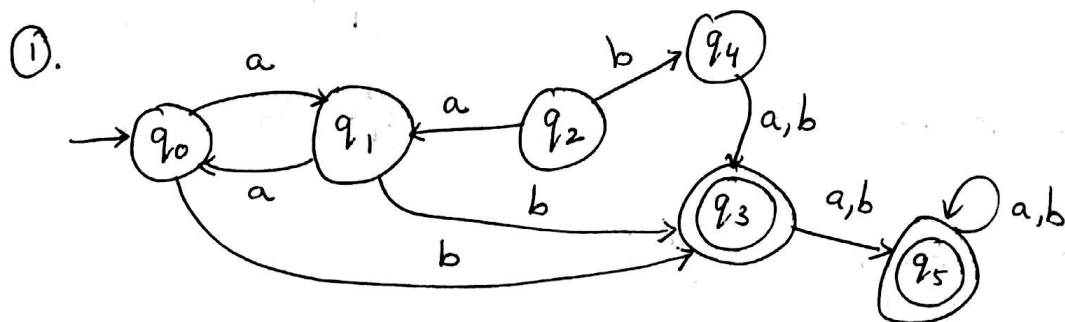
Step 4: If $\Pi_{new} \neq \Pi_{old}$

Copy Π_{new} to Π_{old} .

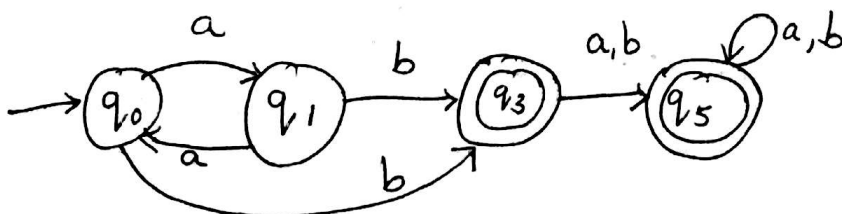
Go to step 3.

Step 5:

From the Π_{new} draw the transition table & DFA.



Remove unreachable states.



Step 0:

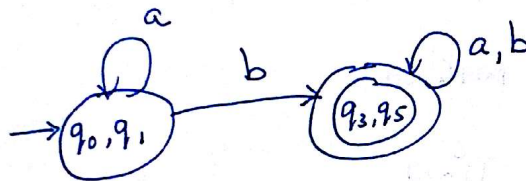
non-final final
 $[q_0 \ q_1] \quad [q_3 \ q_5]$

Step 1:

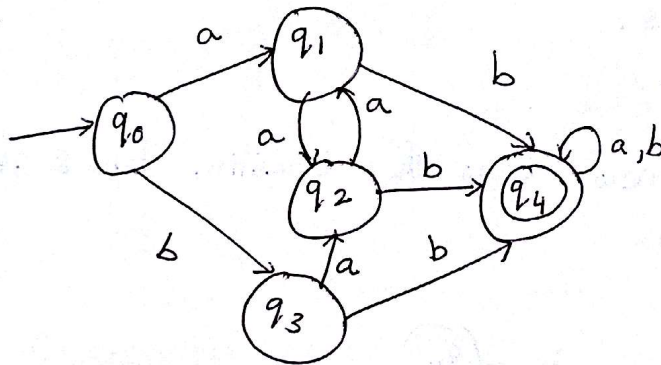
$[q_0 \ q_1] \quad [q_3 \ q_5]$

$\therefore q_0 \equiv q_1$
 $q_3 \equiv q_5$

δ	a	b
q_0, q_1	q_0, q_1	q_3
q_3, q_5	q_5	q_5



②.



No unreachable state.

δ	a	b
$\rightarrow q_0$	q_1	q_3
q_1	q_2	q_4
q_2	q_1	q_4
q_3	q_2	q_4
q_4^*	q_4	q_4

Step 0:

non-final final
 $[q_0 \ q_1 \ q_2 \ q_3] \quad [q_4]$

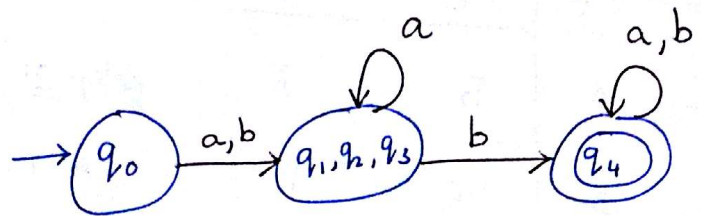
Step 1:

non-fi non-final final
 $[q_0] \quad [q_1 \ q_2 \ q_3] \quad [q_4]$

Step 2:

$[q_0] \quad [q_1 \ q_2 \ q_3] \quad [q_4]$

δ	a	b
$\rightarrow q_0$	q_1	q_3
q_1, q_2, q_3	q_1, q_2	q_4
q_4^*	q_4	q_4



③.

δ	0	1
$\rightarrow A$	B	F
B	G	C^*
C^*	A	C^*
D	C	G
E	H	F
F	C^*	G
G	G	E
H	G	C^*

D is not reachable state
& so it is removed.

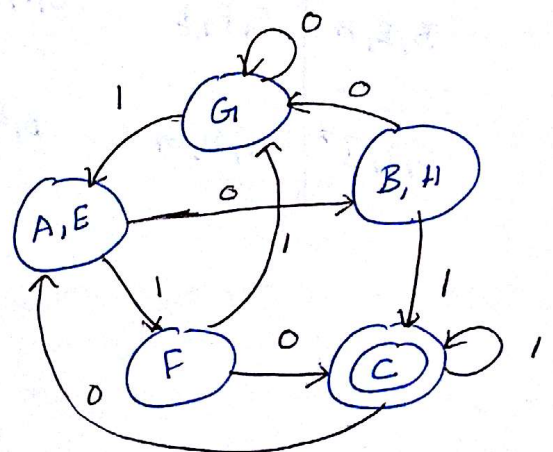
Step 0:
non-final: [A B E F G H] final: [C]

Step 1:
[A E G] [B H] [F] [C]

Step 2:
[A E] [G] [B H] [F] [C]

Step 3:
[A E] [G] [B H] [F] [C]

δ	0	1
A, E	B, H	F
G	G	E
B, H	G	C
F	C	G
C^*	A	C



④

δ	0	1
$\rightarrow A$	B	E
B	C^*	F^*
C^*	D	H
D	E	H
E	F^*	I^*
F^*	G	B
G	H	B
H	I^*	C^*
I^*	A	E

There is no unreachable state

Step 0: non-final [A B D E G H] final [C F I]

Step 1: [A D G] [B E H] [C F I]

Step 2: [A D G] [B E H] [C F I]

$A \equiv D \equiv G$

$B \equiv E \equiv H$

$C \equiv F \equiv I$

δ	0	1
A, D, G	B, E, H	B, E, H
B, E, H	C, F, I	C, F, I
C, F, I*	A, D, G	B, E, H

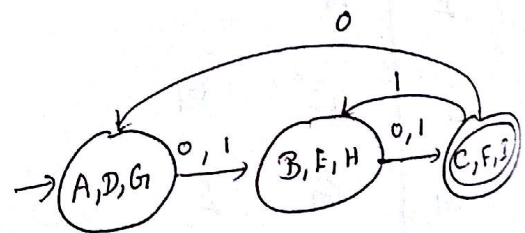


TABLE FILLING METHOD:

①. δ

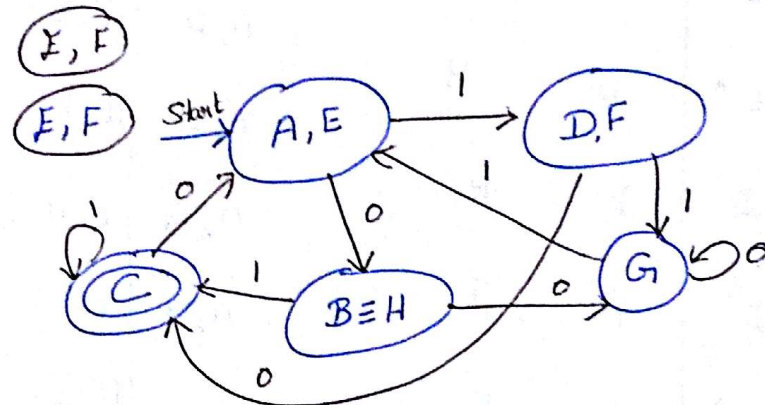
	0	1
$\rightarrow A$	B	F
B	G	C
C*	A	C
D	C	G
E	H	F
F	C	G
G	G	F
H	G	C

B	X						
C	X	X					
D	X	X	X				
E		X	X	X			
F	X	X	X	X	X		
G	X	X	X	X	X	X	
H	X	X	X	X	X	X	X
	A	B	C	D	E	F	G

δ (p,q)	0 (r,s)	1 (r,s)
A,B	B,G	C,F
A,D	B,C	F,G
A,E	B,H	F,F
A,F	B,C	F,G
A,G	B,G	E,F
A,H	B,G	C,F
B,D	C,G	C,G
B,E	G,H	C,F
B,F	C,G	C,G
B,G	G,G	C,E
B,H	G,G	C,C
D,E	C,H	G,F
D,F	C,C	G,G
D,G	C,G	E,G
D,H	C,G	C,G
E,F	C,H	F,G
E,G	G,H	E,F
E,H	H,G	C,F
F,G	C,G	E,G
F,H	C,G	C,G
G,H	G,G	C,E

δ	0	1
✓ A, E	B, H	F, F
✓ D, F	C, C	G, G
✓ B, H	G, G	C, C
E, G	(G, H)	(E, F)
A, G	(B, G)	(E, F)

$A \equiv E$
 $D \equiv F$
 $B \equiv H$

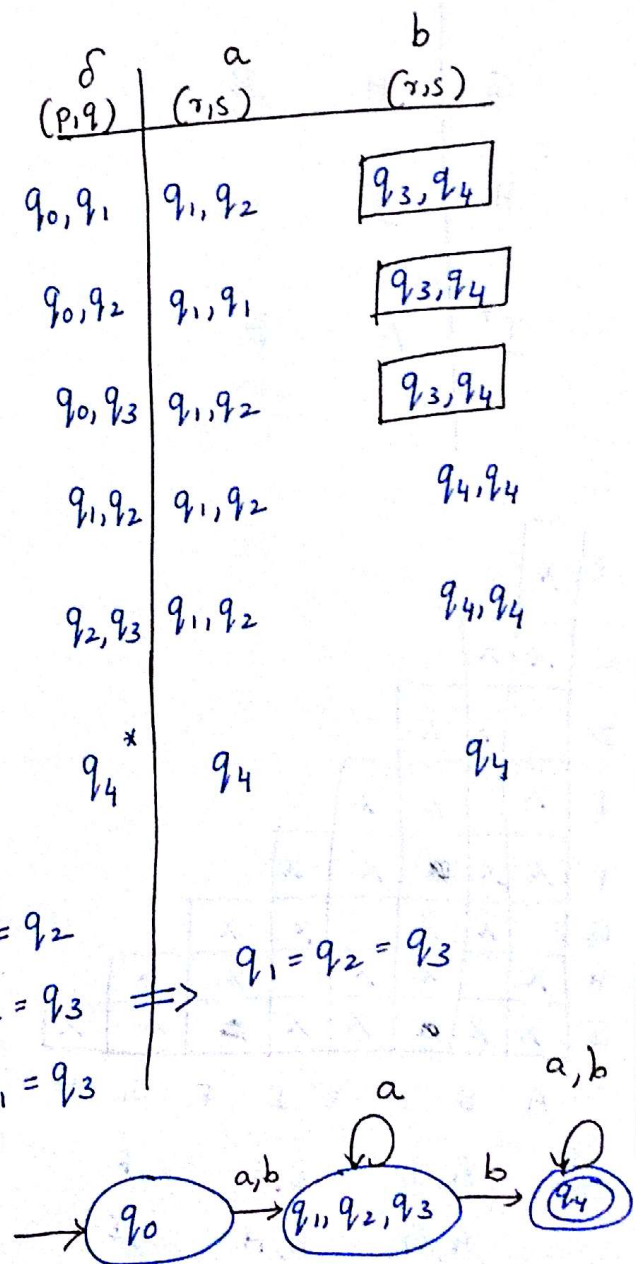


②.

δ	a	b
$\rightarrow q_0$	q_1	q_3
q_1	q_2	q_4
q_2	q_1	q_4
q_3	q_2	q_4
q_4^*	q_4	q_4

q_1	X			
q_2	X			
q_3	X			
q_4	X	X	X	X
	q_0	q_1	q_2	q_3

$$\begin{aligned}
 q_1 &= q_2 \\
 q_2 &= q_3 \\
 q_1 &= q_3
 \end{aligned}
 \Rightarrow q_1 = q_2 = q_3$$



(3)

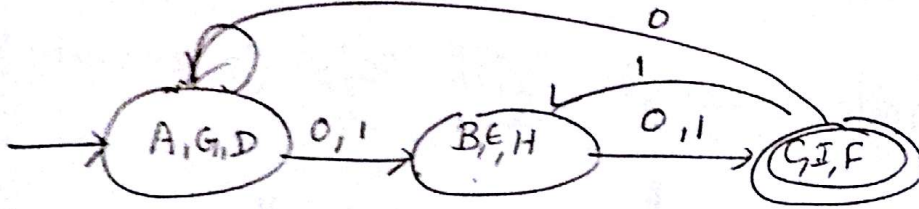
δ	0	1
$\rightarrow A$	B	E
B	C	F
C^*	D	H
D	E	H
E	F	I
F^*	G	B
G	H	B
H	I	C
I^*	A	E

δ	0	1
A, B	<u>B, C</u>	<u>E, F</u>
A, D	B, E	E, H
A, E	<u>B, F</u>	<u>E, I</u>
A, G	B, H	B, E
A, H	<u>B, I</u>	<u>C, E</u>
B, D	<u>C, E</u>	<u>F, H</u>
B, E	C, F	F, I
B, G	<u>C, H</u>	<u>B, F</u>
B, H	C, I	C, F
D, E	<u>E, F</u>	<u>H, I</u>
D, G	E, H	B, H
D, H	<u>E, I</u>	<u>C, H</u>
E, G	<u>F, H</u>	<u>B, I</u>
E, H	F, I	C, I
G, H	<u>H, I</u>	<u>B, C</u>
C, F	D, G	B, H
C, I	A, D	E, H
F, I	A, G	B, E

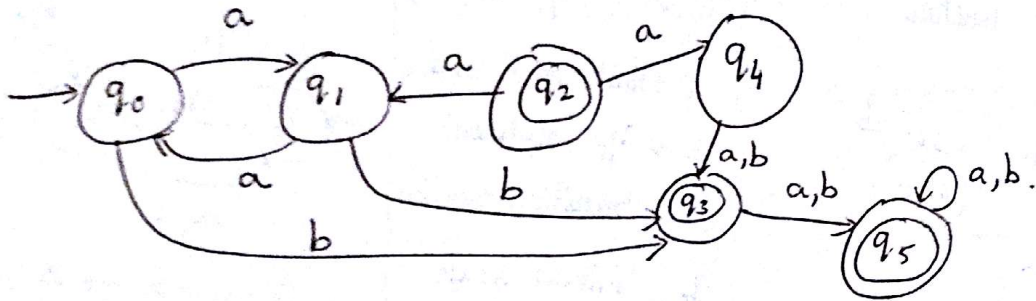
B	X							
C	X	X						
D		X	X					
E	X		X	X				
F	X	X	X	X	X			
G		X	X		X	X		
H	X		X	X		X	X	
I	X	X	X	X	X	X	X	X
	A	B	C	D	E	F	G	H

A, D B, E C, F
 A, G B, H C, I
 D, G E, H F, I

$\Rightarrow A \equiv G \equiv D$ $D \equiv E \equiv H$ $C \equiv I \equiv F$

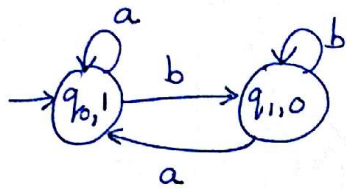


4



Finite Automata with o/p (Deterministic)

Moore machine



$$\lambda: Q \rightarrow \Delta$$

$$q_0 \rightarrow 1$$

$$q_1 \rightarrow 0$$

$$(Q, \Sigma, \delta, q_0, \Delta, \lambda)$$

$Q \rightarrow$ finite set of states

$\Sigma \rightarrow$ i/p alphabet

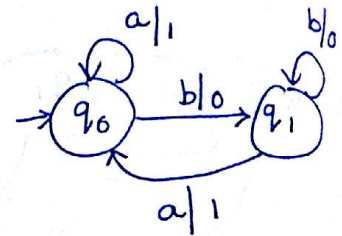
$\delta \rightarrow$ transition function

q_0 initial state

Δ o/p alphabet

$\lambda: \text{o/p function}$

Mealy machine



$$\lambda: Q \times \Sigma \rightarrow \Delta$$

$$(q_0, a) \rightarrow 1$$

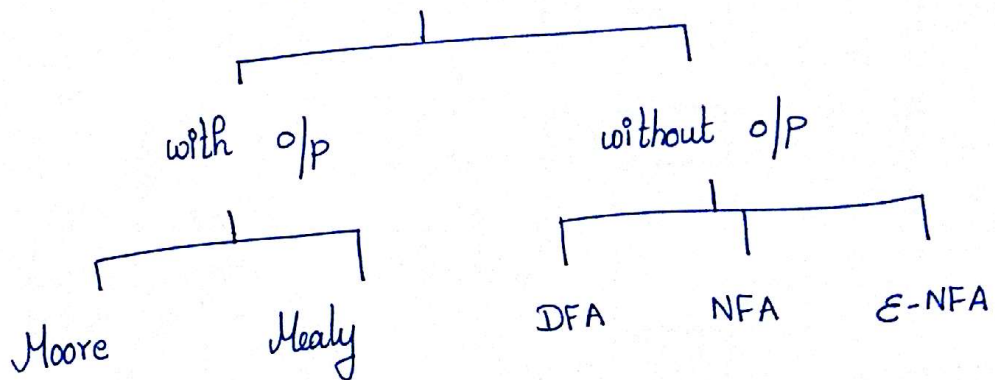
$$(q_0, b) \rightarrow 0$$

$$(q_1, b) \rightarrow 0$$

$$(q_1, a) \rightarrow 1$$

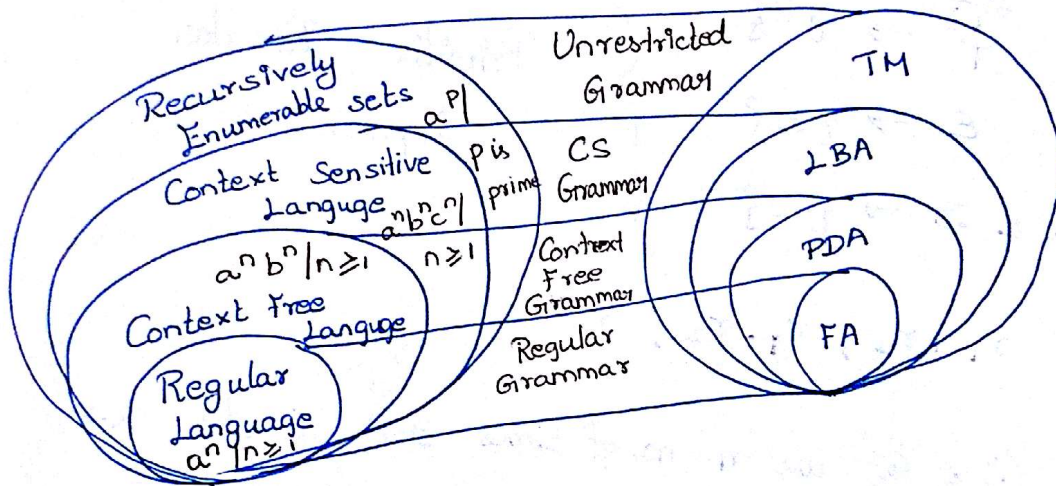
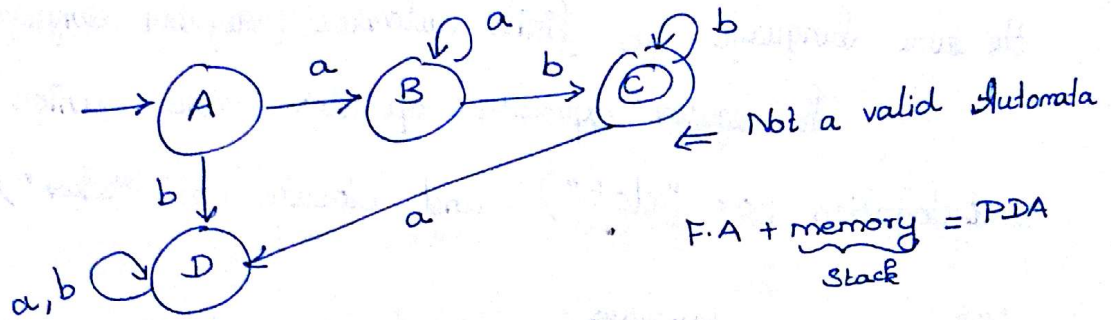
Families of Languages

Finite Automata

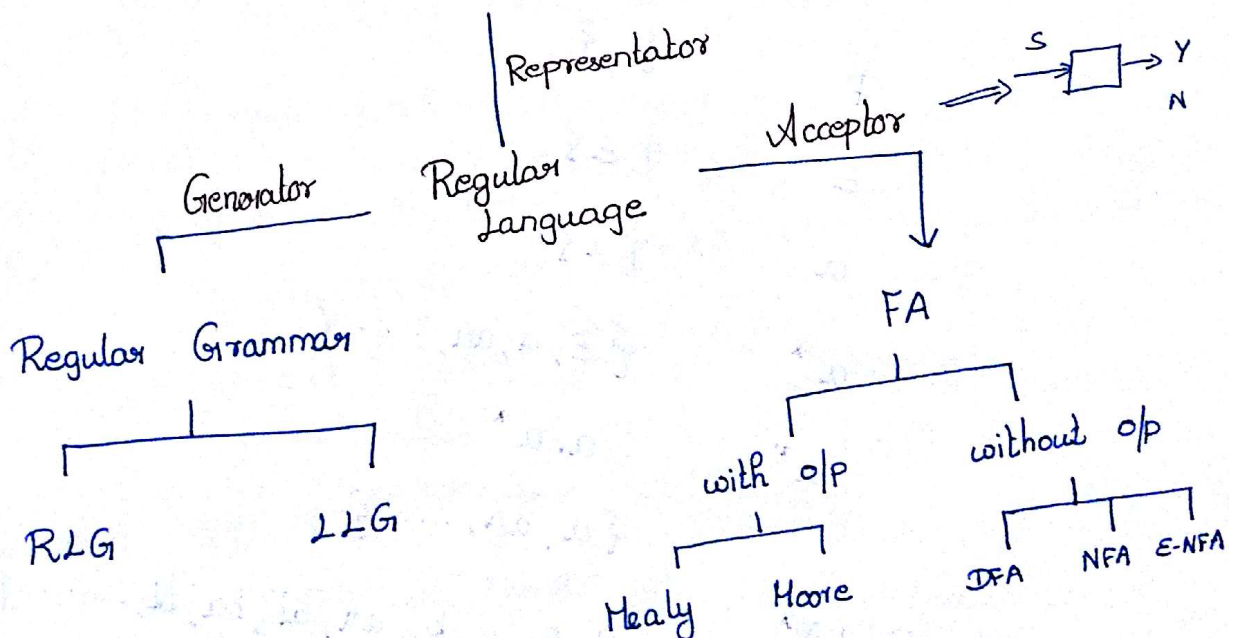


Ex: $L = \{a^n b^n \mid n \geq 1\}$

$L = \{ab, aabb, aaabbb, \dots\}$



Regular Expression.



Regular Expression

It is the algebraic notation that describes exactly the same language as finite automata (regular languages).

The regular expression operators are union, (+) concatenation (or "dot") and closure (or "star").

Note:

	<u>R.E</u>	<u>Language</u>	
①.	Φ	$\{\}$	} Primitive Regular Expressions
	ϵ	$\{\epsilon\}$	
	$a \in \Sigma$	$\{a\}$	

②. $r_1 + r_2, r_1 \cdot r_2, r_1^*$

③. ① & ② use n no. of times that is also R.E

<u>R.E</u>	<u>Language</u>
Φ	$\{\}$
ϵ	$\{\epsilon\}$
a	$\{a\}$
a^*	$\{\epsilon, a, aa, \dots\}$
a^+	$a \cdot a^*$ $\{a, aa, \dots\}$
$(a+b)^*$	$\{\epsilon, a, b, aa, ab, ba, bb, \dots\}$

Examples:

$$\Sigma = \{a, b\}$$

①. $L_1 = \{\text{strings having length exactly } 2\}$

$$= \{aa, ab, ba, bb\}$$

$$= aa + ab + ba + bb$$

$$= a(a+b) + b(a+b)$$

$$= (a+b) \cdot (a+b) \Rightarrow \text{length exactly } 2$$

$$\cancel{= (a+b)(a+b)(a+b)} \Rightarrow \text{length exactly } 3$$

$$(a+b)(a+b)(a+b)(a+b) \Rightarrow \text{length exactly } 4$$

②. $L_2 = \{\text{strings having length at least } 2\}$

$$= \{aa, ab, ba, bb, aaa, \dots\}$$

$$\text{length: } 2, 3, 4, \dots$$



$$(a+b)(a+b)(\underline{a+b})^* \quad \text{0 or more times}$$

③. $L_3 = \{\text{strings having length at most } 2\}$

$$\text{length} = 0, 1, 2.$$

$$= \{\epsilon, a, b, aa, ab, ba, bb\}$$

$$= \epsilon + a + b + aa + ab + ba + bb$$

$$= (\epsilon + a + b)(\epsilon + a + b)$$

④. $L_4 = \{\text{even length strings}\}$

$$L = \{\epsilon, aa, ab, ba, bb, \dots\}$$

$$= ((a+b)(a+b))^*$$

R.E

$$(5). L_5 = \{ \text{strings having odd length} \}$$

$$= ((a+b)(a+b))^* (a+b)$$

$$(6). L_6 = \{ \text{strings length divisible by '3'} \}$$

$$L = 0, 3, 6, 9, 12, \dots$$

$$((a+b)(a+b)(a+b))^*$$

$$(7). L_7 = \{ \text{sets of strings whose length is } \equiv 2 \pmod{3} \}$$

~~is~~

$$(a+b)^{3n+2} / n \geq 0$$

$$((a+b)(a+b)(a+b))^* (a+b)(a+b)$$

$$(8). L_8 = \{ \text{no. of 'a's exactly 2} \} \quad \Sigma = \{a, b\}$$

$$b^* a b^* a b^*$$

$$(9). L_9 = \{ \text{'a's are atleast '2'} \}$$

$$b^* a b^* a (a+b)^*$$

$$(10). L_{10} = \{ \text{'a's almost '2'} \}$$

$$b^* (\epsilon + a) b^* (\epsilon + a) b^*$$

$$(11). L_{11} = \{ \text{set of 'a's are even} \}$$

$$L_{11} = \{ aa, aaaa, \dots, b, bb, bbaa \}$$

$$(b^* a b^* a b^*)^* + b^*$$

$$\equiv (b^* a b^* a)^* \cdot b^*$$

(12). $L_{12} = \{ \text{set of strings starts with } a \}$
 $a(a+b)^*$

(13). $L_{13} = \{ \text{set of strings ends with } a \}$
 $(a+b)^* \cdot a$

(14). $L_{14} = \{ \text{set of all strings containing } a \}$
 $(a+b)^* \cdot a \cdot (a+b)^*$

(15). $L_{15} = \{ \text{starting and ending with diff. symbols} \}$

$$[a(a+b)^*b] + [b(a+b)^*a]$$

(16). $L_{16} = \{ \text{starting and ending with same symbol} \}$
 $L = \{ \epsilon, a, b, aa, bb, \dots \}$

$$[a(a+b)^*a] + [b(a+b)^*b] + [a + b + \epsilon]$$

(17). $L_{17} = \{ \text{2 a's should not come together} \}$

$L = \{ \epsilon, \underline{b}, bb, bbb, a, \underline{a}ba, ab, abab, ababab \}$
 $ba, bab, baa, baba, \dots$

$$(b+ab)^* + (a b + ab)^* \cdot a$$

$$= (b+ab)^* \cdot (\epsilon + a)$$

(or)

$$= (a + \epsilon) \cdot (b + ba)^*$$

(18). $L_{18} = \{ \text{no } a \text{'s \& } b \text{'s come together} \}$
 $= \{ \epsilon, a, b, ab, ba, aba, bab, \dots \}$

	<u>Starts with</u>	<u>Ends with</u>
$\{a, aba, ababa, \dots\}$	a	$a \Rightarrow (ab)^*a$ or $a(ba)^*$
$\{ab, abab, ababab, \dots\}$	a	$b \Rightarrow (ab)^*$ or $a(ba)^*b$
$\{ba, baba, bababa, \dots\}$	b	$a \Rightarrow (ba)^*$ or $b(ab)^*a$
$\{b, bab, babab, bababab, \dots\}$	b	$b \Rightarrow (ba)^*b$ or $b(ab)^*$

$$(ab)^*a + (ab)^* + b(ab)^*a + b(ab)^*$$

$$= (ab)^*(a + \epsilon) + [b(ab)^*](a + \epsilon)$$

$$= (\epsilon + b) \underset{(or)}{(ab)^*} (a + \epsilon)$$

$$(\epsilon + a) (ba)^* (b + \epsilon)$$

Identities of Regular Expressions:

Empty set $\rightarrow \emptyset \in \mathcal{R}^L$ (or) $\mathcal{R}^L$

$$(*) \quad \emptyset + R = R + \emptyset = R$$

$$(*) \quad \emptyset \cdot R = R \cdot \emptyset = \emptyset$$

$$(*) \quad \epsilon \cdot R = R \cdot \epsilon = R$$

$$(*) \quad \epsilon^* = \epsilon$$

$$(*) \quad \emptyset^* = \epsilon$$

$$(*) \quad \epsilon + R \cdot R^* = R^* R + \epsilon = R^*$$

$$\begin{aligned} (*) \quad (a+b)^* &= (a^* + b^*)^* \\ &= (a^* b^*)^* \\ &= (a^* + b)^* \\ &= (a + b^*)^* \\ &= a^* (b a^*)^* \\ &= b^* (a b^*)^* \end{aligned}$$

Kleen's Theorem

Part ①: Any regular language can be accepted by a FA.

②: The language accepted by any finite automaton is regular.

$$(*) \quad L + M = M + L$$

$$(*) \quad (L + M) + N = L + (M + N)$$

$$(*) \quad (LM) \cdot N = L(MN)$$

$$(*) \quad L(M + N) = LM + LN$$

$$(*) \quad (M + N)L = M \cdot L + NL$$

$$(*) \quad L + L = L$$

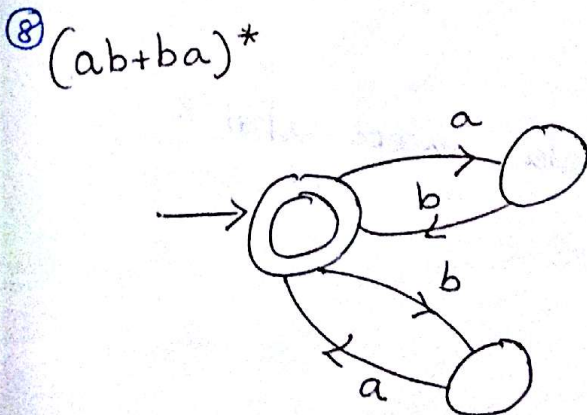
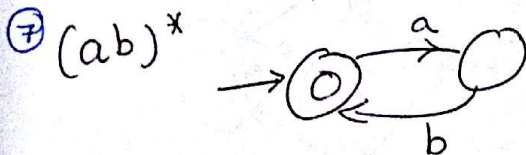
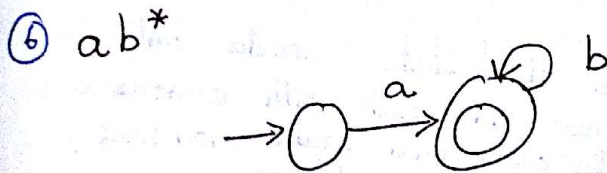
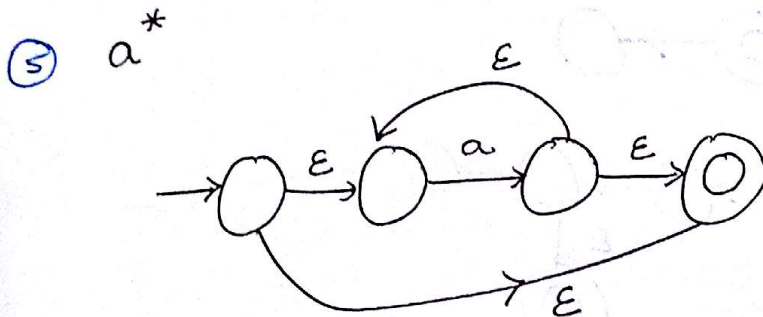
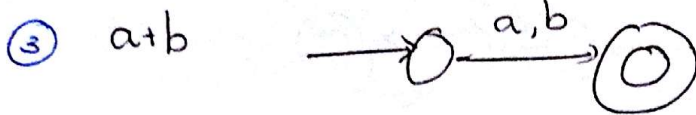
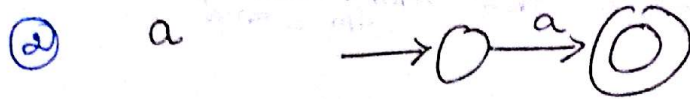
$$(*) \quad (L^*)^* = L^*$$

$$(*) \quad L^+ = LL^* = L^*L$$

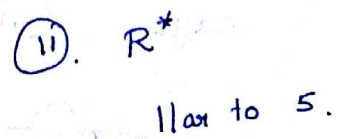
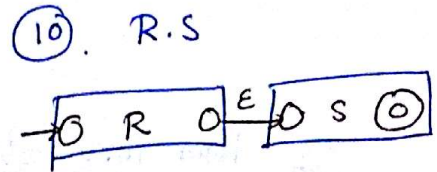
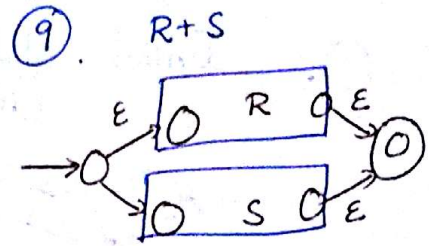
$$(*) \quad L^* = L^+ + \epsilon$$

$$(*) \quad (L + M)^* = (L^*M^*)^*$$

Regular Expression to FA



RE $\rightarrow$ E-NFA

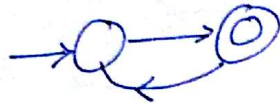


|| or to S.

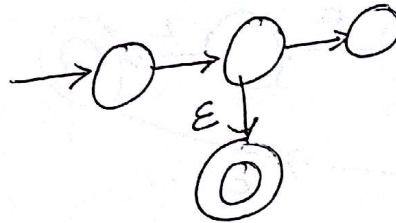
Finite Automata to Regular Expression

State Elimination Method:

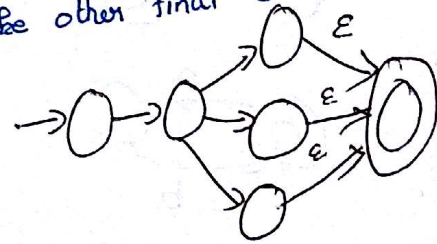
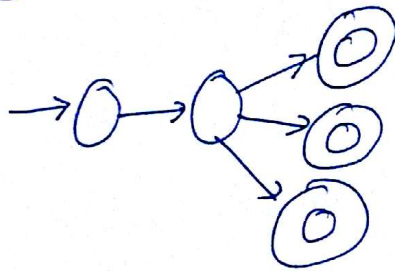
- ①. Initial state should not have incoming edge
(if so create new initial state with ϵ move)



- ②. From final state there should not be any outgoing edge
(if so create final state with ϵ moves) and make old final state as non-final

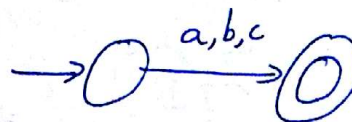
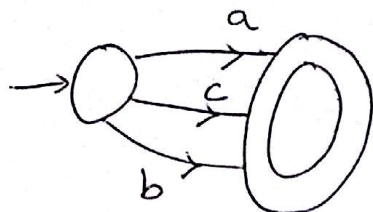


- ③. If there is more than one final state create only one new final state with ϵ -moves & make other final states non-final.



- ③. Eliminate all the other states except start & final states.

⊛₁



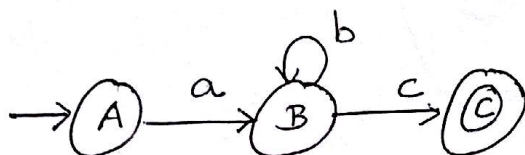
$a+b+c$

⊛₂



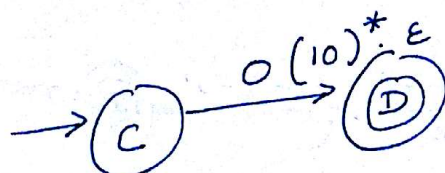
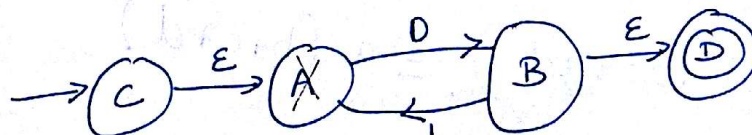
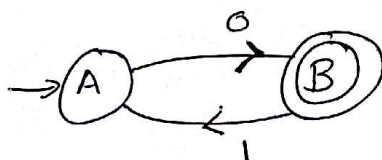
ab

⊛₃



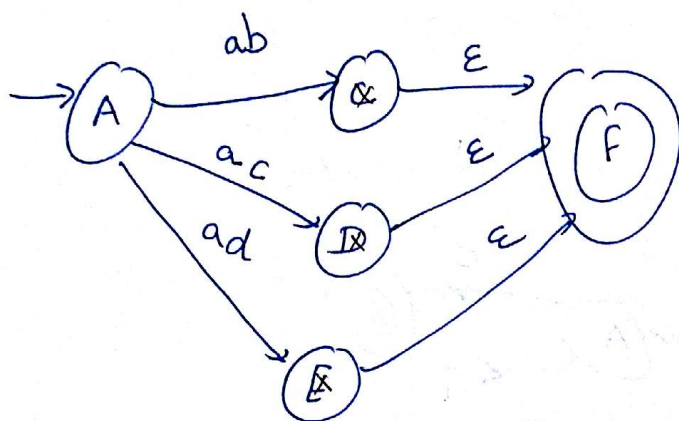
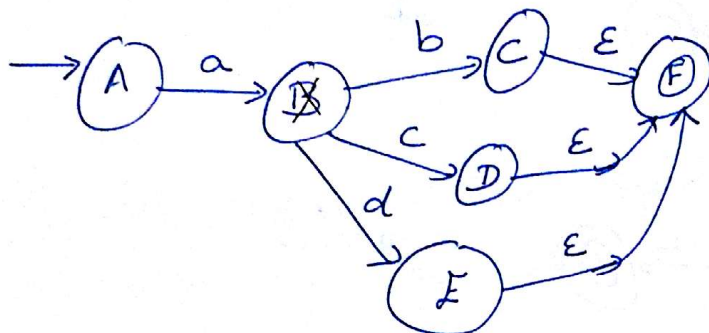
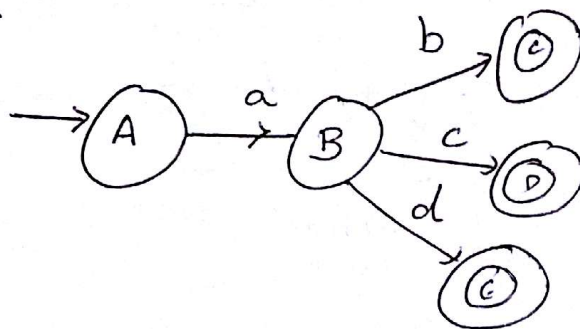
ab^*c

⊛₄



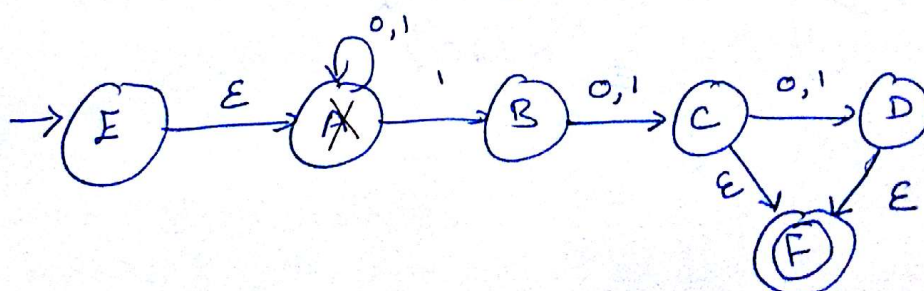
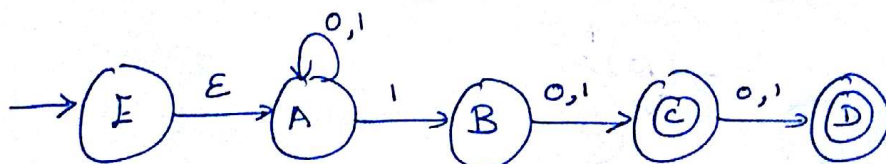
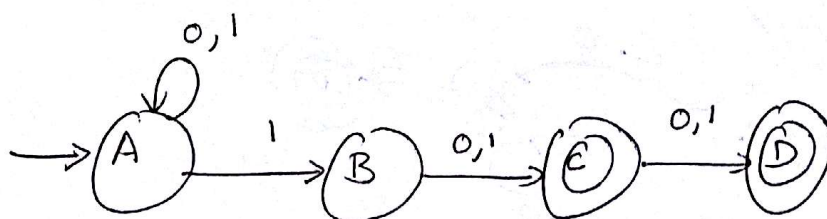
$\equiv 0(10)^*$

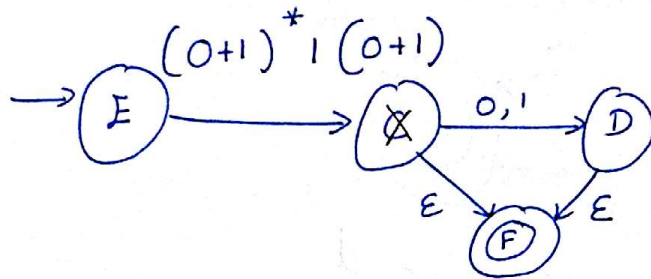
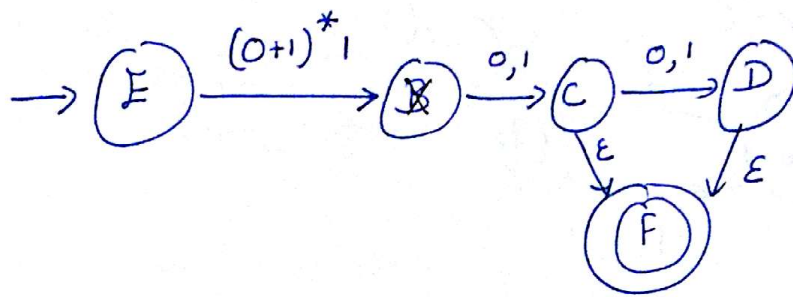
*5



$$(ab + ac + ad) = a(b + c + d)$$

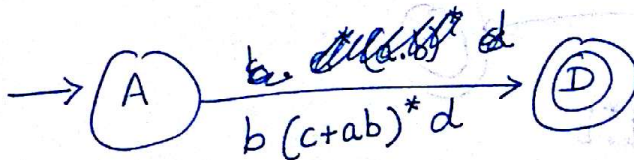
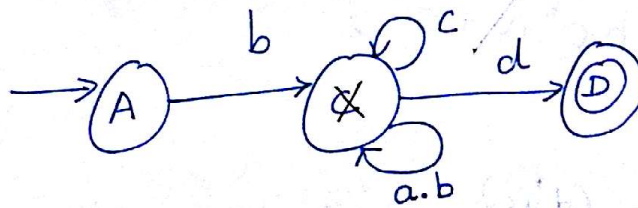
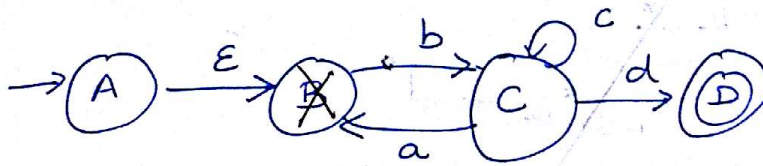
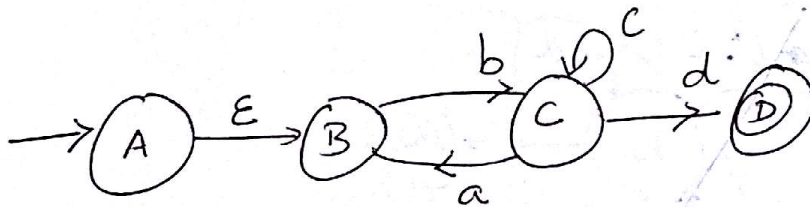
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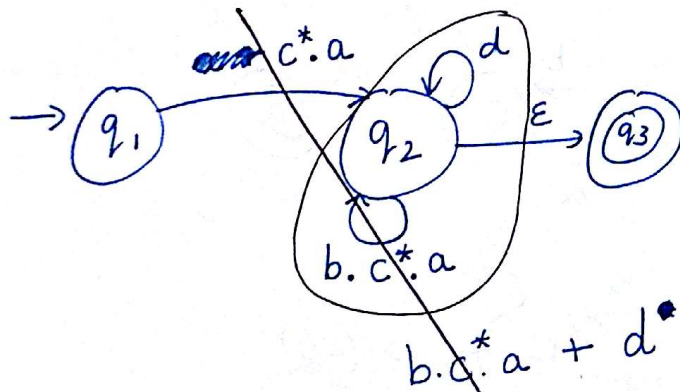
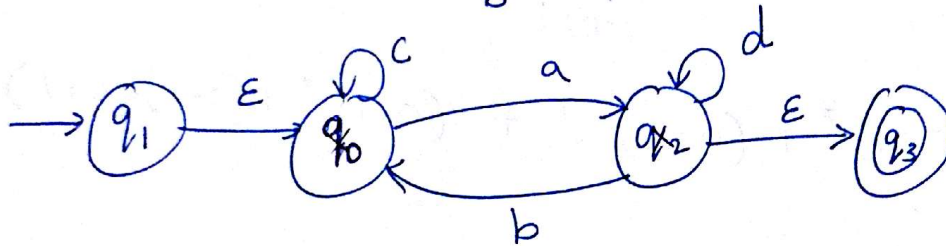
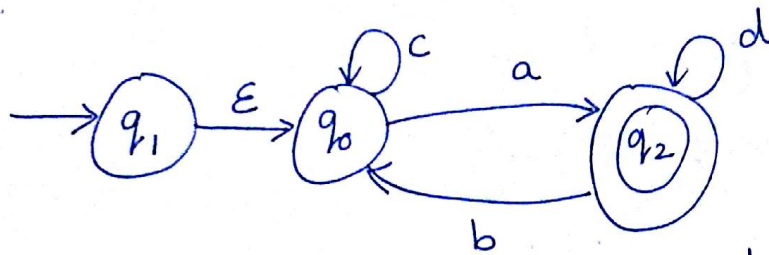
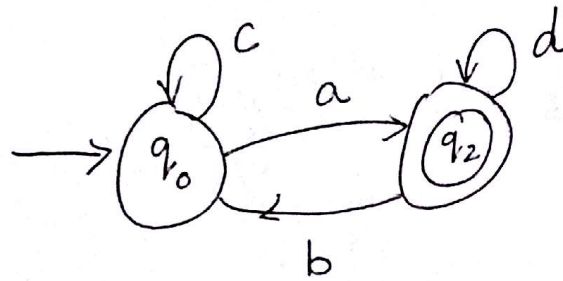
$$(0+1)^* 1 (0+1) + (0+1)^* 1 (0+1) (0+1)$$

Q7.

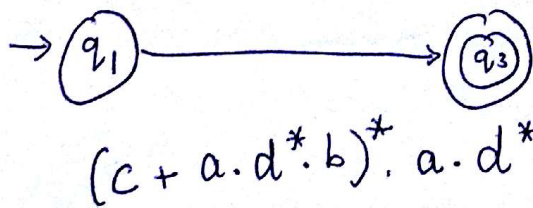
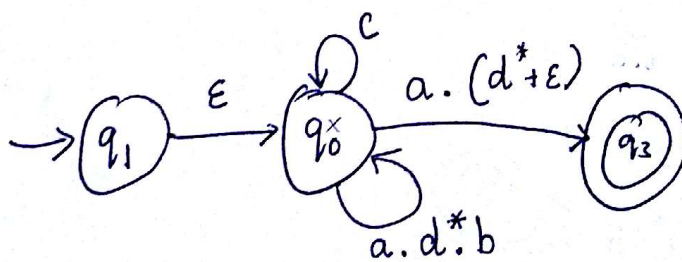


$$b(c+ab)^* d$$

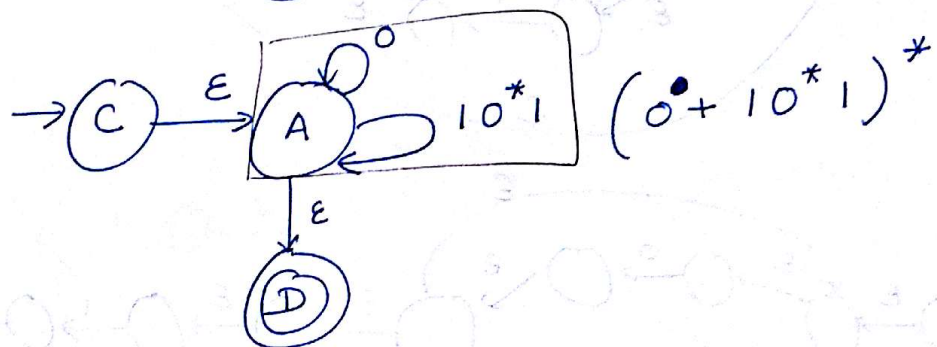
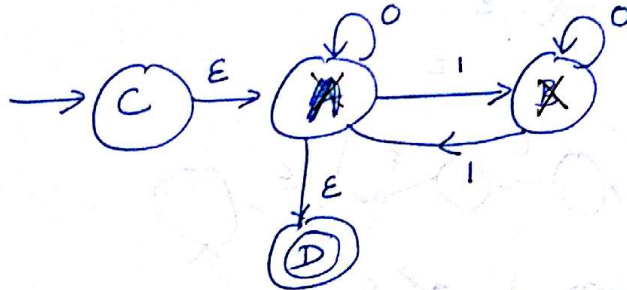
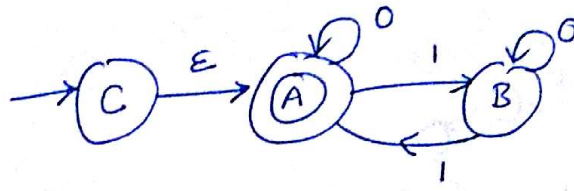
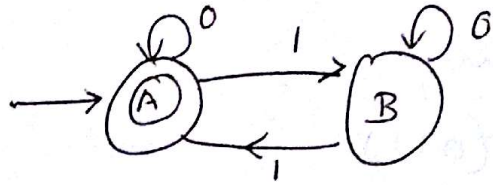
*8



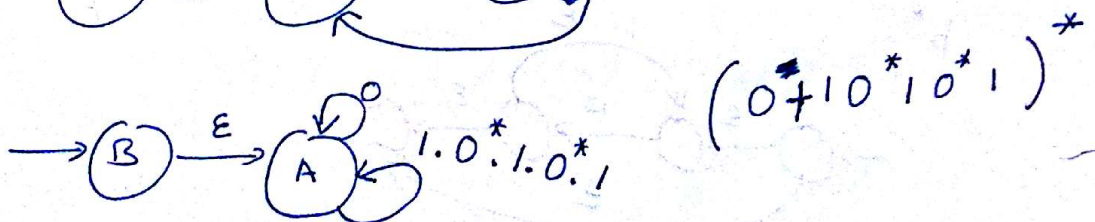
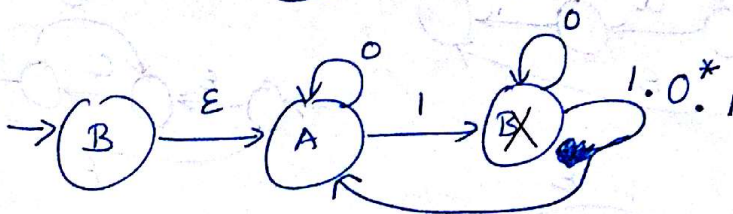
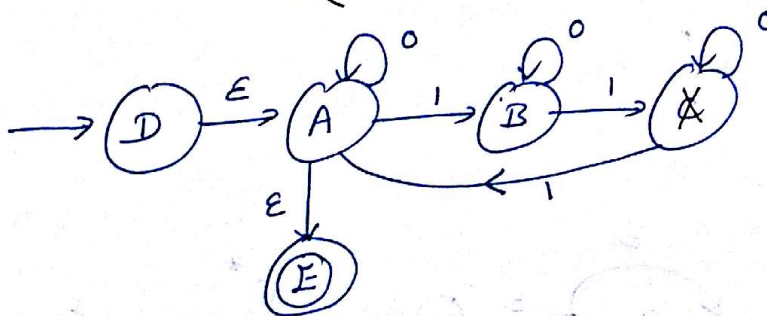
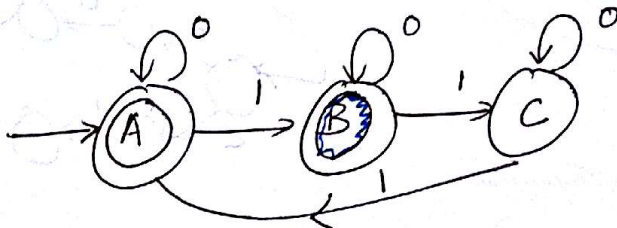
$$[c^*.a] \cdot (b.c^*.a + d^*)^*$$



*9



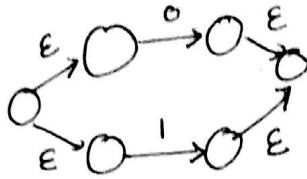
*10



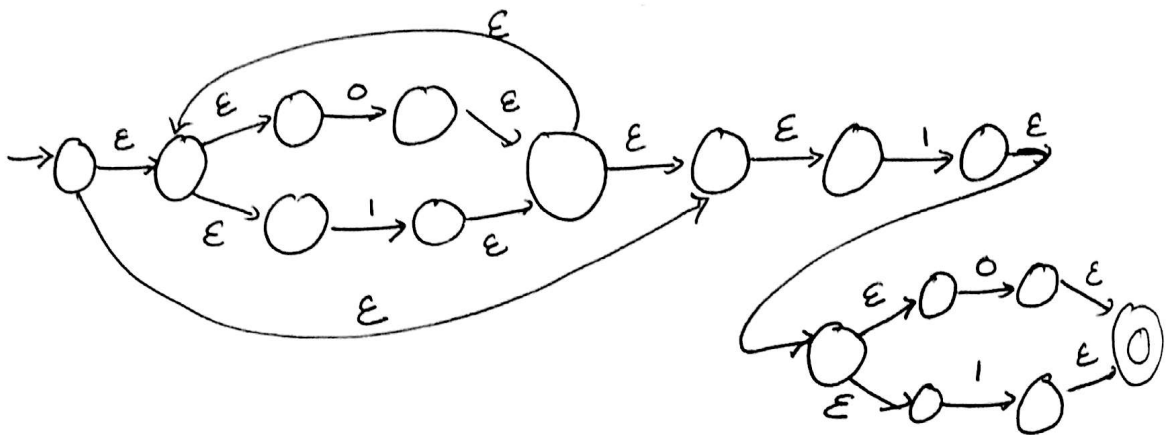
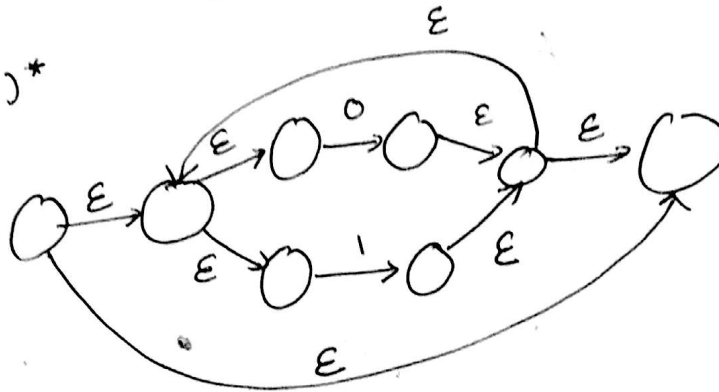
Convert the R.E to E-NFA:

① $(0+1)^* 1 (0+1)$

$0+1$

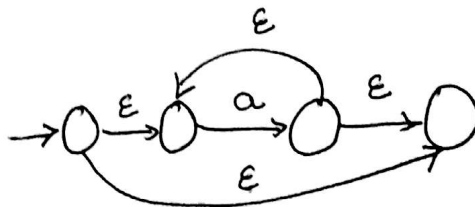


$(0+1)^*$

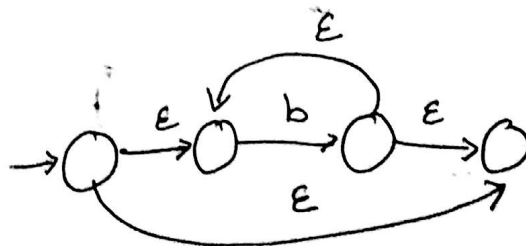


② $a^* + b^* + c^*$

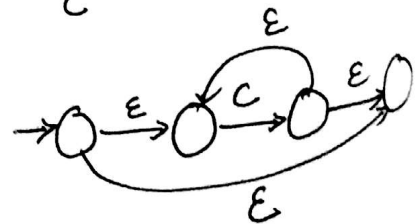
a^*

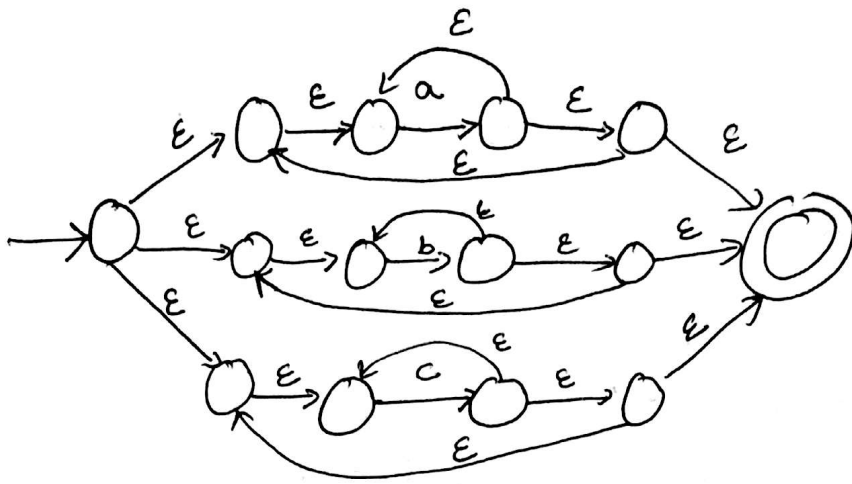


b^*

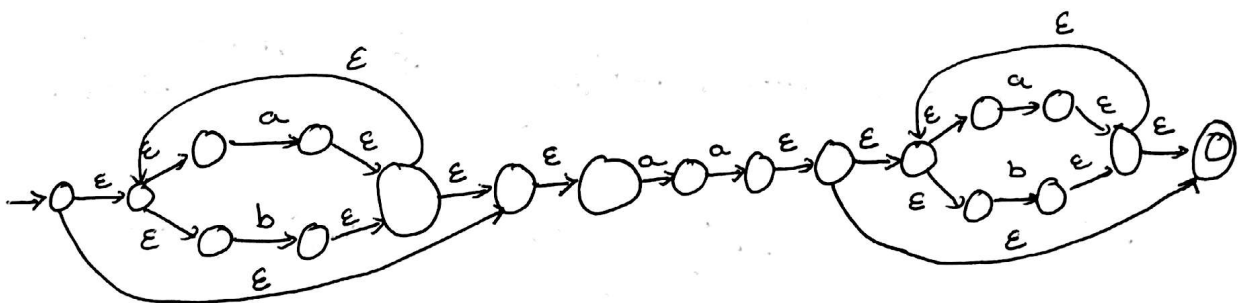
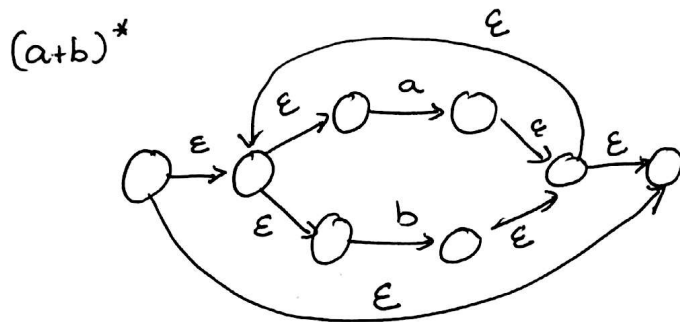


c^*

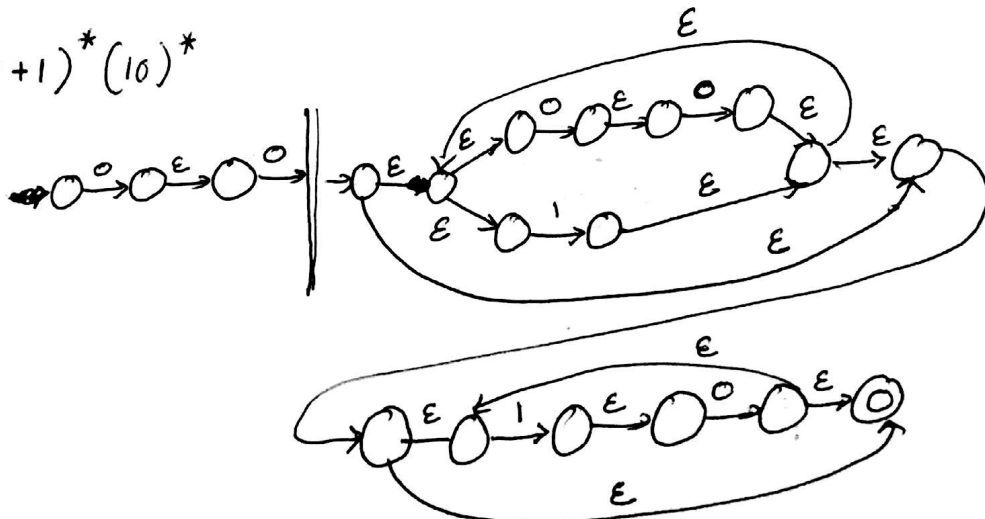




③. $(a+b)^* aa(a+b)^*$



④. $(00+1)^* (10)^*$



Kleene's Theorem Part 2

The language accepted by any finite automata is regular. $\boxed{FA \rightarrow R \cdot \Sigma}$

$$A = (Q, \Sigma, \delta, q, F)$$

$$Q = \{1, 2, 3, \dots, n\}$$

$R_{ij}^k = \{w \in \Sigma^* \mid w \text{ is label from } i \text{ to state } j \text{ that goes through an intermediate state whose no. is not greater than } k\}$

$$w = xy$$

$$\hat{\delta}(i, x) = k \quad \hat{\delta}(k, y) = j$$

$$\boxed{k=0}$$

$$i \rightarrow j$$

$$\textcircled{1}. i \neq j$$

$$\textcircled{a}. R_{ij}^0 = \phi \text{ (no } i/p)$$

$$\textcircled{b}. R_{ij}^0 = a \text{ (one } i/p)$$

$$\textcircled{c}. R_{ij}^0 = a_1 + a_2 + \dots + a_m \text{ (multi } i/p)$$

$$\textcircled{2}. i = j$$

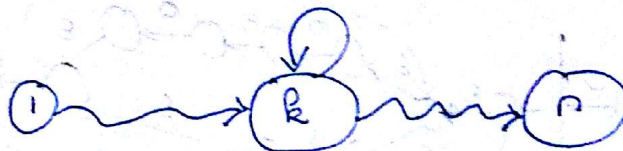
$$\textcircled{a}. R_{ij}^0 = \phi + \epsilon \text{ (no } i/p)$$

$$R_{ij}^0 = a + \epsilon \text{ (one } i/p)$$

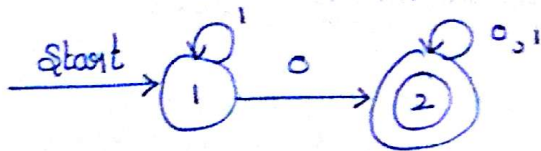
$$R_{ij}^0 = a_1 + a_2 + \dots + a_m \text{ (multi } i/p)$$

$$\boxed{k \neq 0}$$

$$R_{ij}^k = R_{ij}^{k-1} + R_{ik}^{(k-1)} (R_{kk}^{(k-1)})^* R_{kj}^{(k-1)}$$



①.

 $k = 0, 1, 2.$ $k=0$

$$R_{11}^0 = 1 + E$$

$$R_{12}^0 = 0$$

$$R_{21}^0 = \phi$$

$$R_{22}^0 = 0 + 1 + E$$

 $k=1$

$$R_{ij}^k = R_{ij}^{k-1} + R_{ik}^{k-1} (R_{kk}^{k-1})^* R_{kj}^{k-1}$$

$$R_{11}^1 = R_{11}^0 + R_{11}^0 (R_{11}^0)^* R_{11}^0$$

$$\begin{aligned} R_{11}^1 &= (1+E) + (1+E)(1+E)^* (1+E) \\ &= (1+E)(1+E)^* (1+E) \\ &= 1^* \end{aligned}$$

$$\begin{aligned} R_{12}^1 &= R_{12}^0 + R_{11}^0 (R_{11}^0)^* R_{12}^0 \\ &= 0 + (1+E)(1+E)^* \cdot 0 \\ &= 0 + 1^* \cdot 0 = 1^* \cdot 0 \end{aligned}$$

$$\begin{aligned} R_{21}^1 &= R_{21}^0 + R_{21}^0 (R_{11}^0)^* R_{12}^0 \\ &= \phi + \phi (1+E)^* (1+E) \\ &= \phi \end{aligned}$$

$$\begin{aligned} R_{22}^1 &= R_{22}^0 + R_{21}^0 (R_{11}^0)^* R_{12}^0 \\ &= (0+1+E) + \phi \cdot (1+E)^* \cdot 0 \\ &= 0+1+E \end{aligned}$$

 $k=2$

$$\begin{aligned} R_{11}^2 &= R_{11}^1 + R_{11}^1 (R_{11}^1)^* R_{11}^1 = (1+E)^* (1+E)^* (1+E)^* (1+E) \\ &= 1^* + 1^* \cdot 0 (0+1+E) \cdot \phi = 1^* \end{aligned}$$

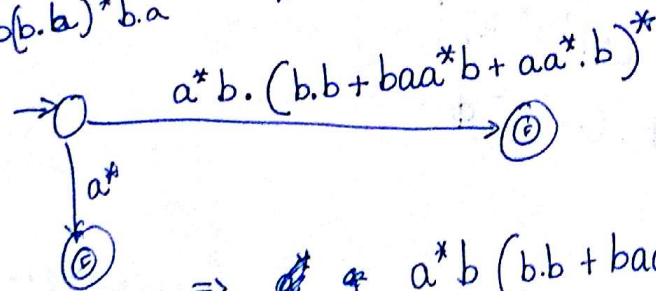
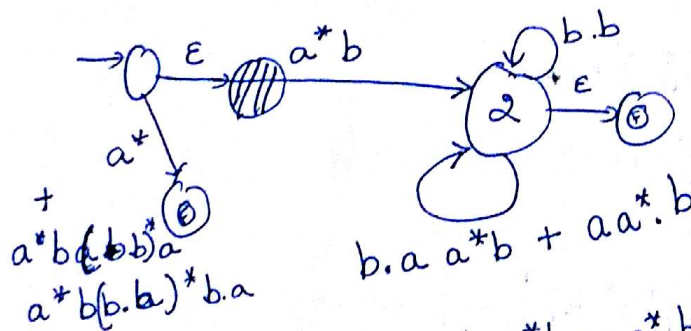
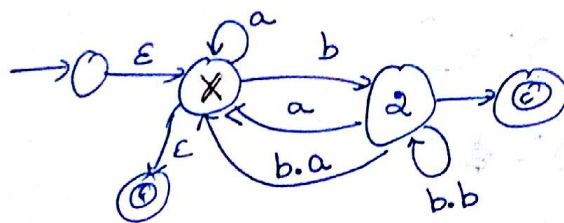
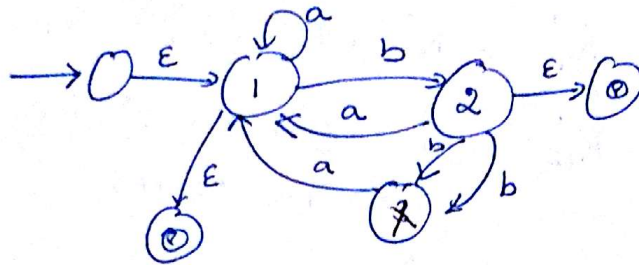
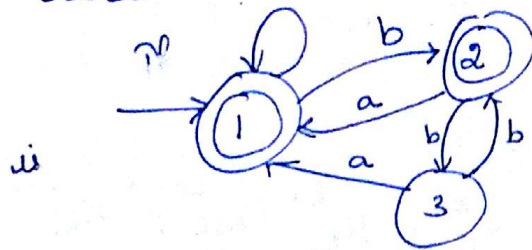
$$R_{12}^2 = R_{12}^1 + R_{11}^1 (R_{11}^1)^* R_{12}^1 = 1^* \cdot 0 + 1^* \cdot 0 (0+1+E)^* (0+1+E)$$

$R_{12}^2 = 1^* \cdot 0 (0+1)^*$

$$\begin{aligned} R_{21}^2 &= R_{21}^1 + R_{21}^1 (R_{11}^1)^* R_{12}^1 = \phi + (0+1+E) \cdot (0+1+E)^* \cdot \phi \\ &= \phi \end{aligned}$$

$$\begin{aligned} R_{22}^2 &= R_{22}^1 + R_{21}^1 (R_{11}^1)^* R_{12}^1 = (0+1+E) + (0+1+E)(0+1+E)^* (0+1+E) \\ &= (0+1)^* \end{aligned}$$

Kleene's $\mathcal{N}$ State Elimination Method

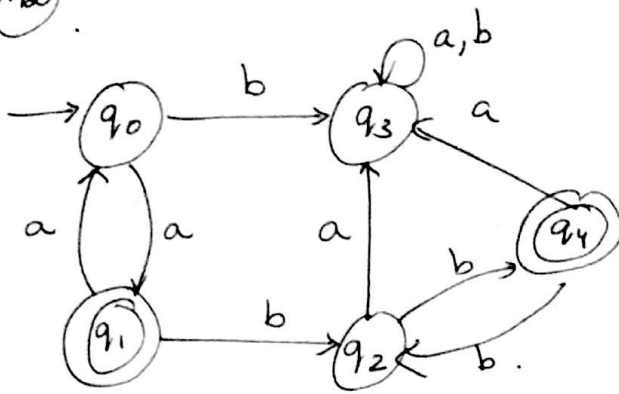


$$\Rightarrow a^*b(b.b + baa^*b + aa^*.b)^*$$

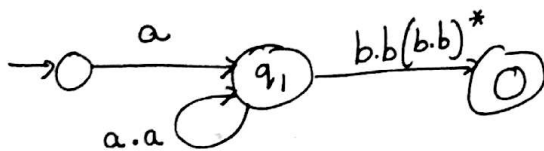
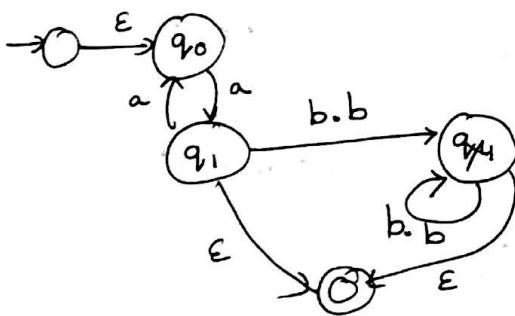
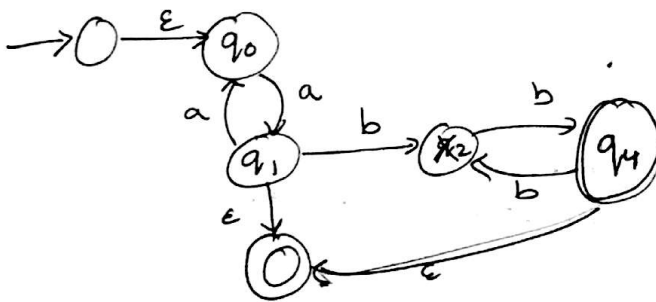
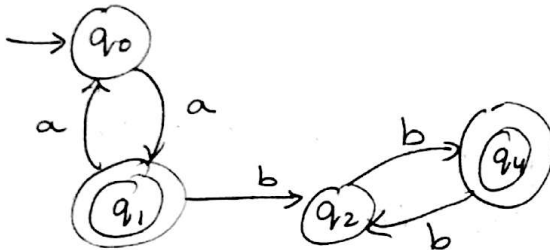
$$a^* + a^*ba + a^*b.b.a$$

$$a^* + a^*b(bb)^*a + a^*b(bb)^*b.a$$

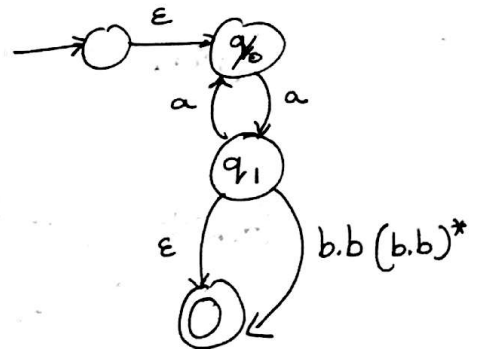
X12



Remove non-reachable state q_3 .



$$(aa)^* a + a(a.a)^* b.b(b.b)^*$$



Pumping Lemma for Regular Languages:

Statement:

Let L be a regular language. There exists a constant n such that ~~any sentence~~ for every string w in L such that $|w| \geq n$, we can break w into three strings, $w = xyz$, such that:

$$y \neq \epsilon \Rightarrow |y| \geq 1$$

$$|xy| \leq n$$

For all $k \geq 0$, the string xy^kz is also in L .

⊗ $y \Rightarrow$ if it is pumped any number of times or deleting it keeps the resulting string in the language L .

Proof:

Assume L is regular.

$\Rightarrow L = L(A)$ for some DFA A .

$$|A| = n$$

String w $|w| = n$ or more.

$$w = a_1 a_2 \dots a_m \quad (m \geq n)$$

Transition function of A

$$i = 0, 1, \dots, n$$

$$\delta(q_0, a_1 a_2 \dots a_i)$$

Break $w = xyz$ as follows:

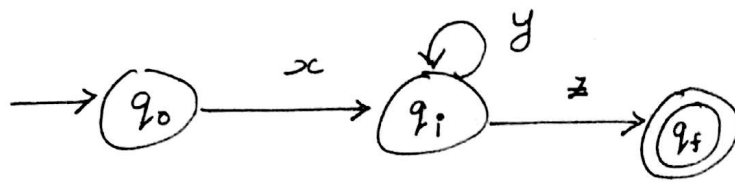
$$x = a_1 a_2 \dots a_i$$

$$y = a_{i+1} a_{i+2} \dots a_j$$

$$0 \leq i < j \leq n.$$

$$z = a_{j+1} a_{j+2} \dots a_m.$$

[x may be empty; z may be empty.
 y cannot be empty, since i is strictly less than j]



[Every string longer than the number of states must cause a state to repeat]

- (*)₁ If $k=0$ then xz will be accepted.
- (*)₂ If $k>0$ then xy^kz will be accepted.

Pumping lemma can be applied only to infinite languages.

Applications:

- (*)₁ It is mainly used to show that certain languages are not regular.

(*)₂ Points:

Language is Finite - Regular

Infinite $\begin{cases} \text{Regular} \Rightarrow \text{Some pattern} \Rightarrow \text{No FA is possible} \\ \text{Not Regular.} \end{cases}$

Pumping Lemma $\begin{cases} \rightarrow \text{Regular or not regular} \\ \rightarrow \text{Not Regular (Negativity Test)} \end{cases}$

①. Show that $L = \{a^n b^n \mid n \geq 0\}$ is not regular.

①. Let L is regular and n be the number of states in FA.

$$\text{String } w = a^n b^n$$

②. $|w| = 2n \geq n$

$$w = xyz$$

such that $|xy| = n$ $|y| = 1$.

$$w = \underbrace{aaaa \dots a}_x \underbrace{a}_{y} \underbrace{bb \dots b}_z$$

According to pumping lemma.

$$xy^kz \in L \text{ for } k=0, 1, 2, \dots$$

③. $k=0$.

y does not appear.

$\Rightarrow$ $n-1$ a's are followed by n b's $\Rightarrow$ is not in the language.

By contradiction to the assumption that the language is regular.

So, the language $L = \{a^n b^n \mid n \geq 0\}$ is not regular.

②. Show that $L = \{ww^R \mid w \in (0+1)^*\}$ is not regular.

①. Let L is regular and n be the number of states in FA.

$$\text{String } w = \underbrace{1 \dots 1}_n \underbrace{00}_n \underbrace{00 \dots 0}_n \underbrace{1}_{n^R}$$

total

(ii). $|w| = 4 \cdot n \geq n$.

$w = xyz$

such that $|xy| = n$ $|y| = 1$

$w = \underbrace{1 \dots 1}_{xy} \underbrace{0 \dots 0}_{z} 0 \dots 0 1 \dots 0$

$x = 1 \dots 1 \Rightarrow |x| = n-1$
 $|y| = 1$

According to pumping lemma.,

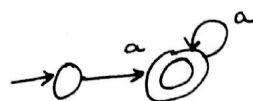
$xy^kz \in L$ for $k=0,1,2,\dots$

(iii). $k=0$ y does not appear $\Rightarrow$ no. of 1's appear in LHS will be $<$ than RHS.

So $xy^kz \notin L$ when $k=0$, which is a contradiction to the assumption.

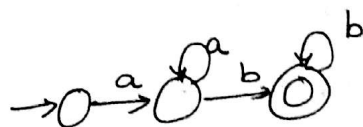
$\therefore L = \{w w^R \mid w \in \{0+1\}^*\}$ is not regular.

(3). $L_1 = \{a^n \mid n \geq 1\}$



Regular

$L_2 = \{a^n b^m \mid n, m \geq 1\}$



Regular.

$L_3 = \{a^n b^n \mid n \leq 10^{10^{10}}\}$

$\Rightarrow n$ is bounded so language is finite & so L_3 is Regular

$L_4 = \{w w^R \mid |w| = 2, Z = \{a, b\}\}$

$\Rightarrow$ language is finite & so L_4 is Regular

$L_5 = \{a^n b^m c^k \mid n, m, k \geq 1\}$

$\Rightarrow$ Regular. (RE $aa^*bb^*cc^*$)

$L_6 = \{a^i b^{2j} \mid i, j \geq 1\}$

$\Rightarrow$ Regular. $(aa^*bb(bb)^*)$

$L_7 = \{a^i b^{4j} \mid i, j \geq 1\}$

$\Rightarrow$ Regular. $(aa^*bbbb(bbbb)^*)$

not in
language.
the

regular.

regular.

states

④. $\Sigma = \{a\}$

Arth. progression

Common diff = 2.

$a^n \mid n \text{ is even} \Rightarrow \{a^0, a^2, a^4, a^6, a^8, \dots\}$

$a^n \mid n \text{ is odd} \Rightarrow \{a, a^3, a^5, a^7, \dots\}$ A.P.

$a^n \mid n \text{ is prime} \Rightarrow$ X

$a^{n^2} \mid n \geq 1 \Rightarrow \{a, a^4, a^9, a^{16}, \dots\}$ X A.P.

$a^{2^n} \mid n \geq 1 \Rightarrow \{a, a^2, a^4, a^8, a^{16}, \dots\}$ X A.P.

⑤. Show that $L = \{0^n \mid n \text{ is prime}\}$ is not regular.

①. Let L is regular & n be the number of states in FA.

String $w = 0^n$.

②. $|w| = n$.

$$w = 0^n = \underbrace{0^j}_x \underbrace{0^k}_y \underbrace{0^{n-j-k}}_z$$

$|x| = j$

$|y| = k \geq 1$

$|x \cdot y| = |x| + |y| = j + k \leq n$.

③. $j, k = 0$.

$0^j (0^k)^i 0^{n-j-k} \in L$.

$j + ki + n - j - k = n + k(i-1)$ is prime for all $i \geq 0$.

if $i = n+1$

then $n + k(i-1) = n + kn = n(k+1)$ is also prime
for each $k \geq 1$.

which is contradiction to the assumption that the
language is regular.

So $L = \{0^n \mid n \text{ is prime}\}$ is not regular.